

A SKILL DOCUMENT’S EFFECT LIVES ON ITS OWN TOKENS

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ABSTRACT

A short skill document—a procedure, a lookup table, a worked example—placed in a language model’s context reliably improves accuracy. Where that improvement lives inside the model is not known, and the standard place to look for it is empty: on 55 third-party clinical-calculator skills, every vector written into the last prompt position lands on the no-document baseline, including replacing the state outright. Changing only the answer format, a single-position patch recovers 1.000 of the rescued items for a one-token answer, 0.30 for a short number and **0.000** under a chain of thought: that position carries the model’s *decision*, and a decision spread over hundreds of tokens does not fit. The channel that does carry is the document’s own token span, and because a length-matched receiver already holds a document, substituting it is content injection at $\alpha = 1$ over m positions—which recovers 0.87 of the document’s effect on forty calculators while half a dose, another skill, permuted positions and noise all stay at or below 0.10. Truncating the injected object’s rank measures how large it is: 0.13 at rank 32, 0.77 at 128, nothing for the discarded bottom 64 directions. Within a model family that number does not scale with width ($k^* = 40$ on a model four times narrower than the one giving 52, against a pre-registered width-proportional 12), but it is not a property of the document alone: Mistral-7B, at the same width, needs 64–84. The channel, its controls and a dissociation between where the document can be handed over and where it is still read replicate on a second task domain and a second model family. Along the way we retract a mechanism of our own: an apparent collapse of the span’s participation ratio is massive activations, not compression.

1 INTRODUCTION

A “skill” is a short document—a procedure, a formula, a lookup table, a worked example—placed in a model’s context so it can do a task it otherwise cannot. A benchmark literature measures its effect; none of those measurements says what the document does inside the model.

The natural way to look is to compare the internal state with and without the document, and it does not work in its obvious form. At the last prompt position the residual stream moves a long way when a skill is added—and, through the layers in which the document is actually read, almost exactly as far and in almost the same direction when the document is replaced by an unrelated one of the same length (cosine 0.91–0.95 from layer 0 to 10). Read naively that says the displacement carries little about what the document says. Read against behaviour it cannot: on 1,100 clinical-calculator instances with third-party skills the gold skill is worth +39.7pp over no document while a length-matched *wrong* skill is worth −2.9pp [−8.8, +3.0]. Behaviour is all content; geometry, over the layers that do the reading, is almost all presence.

Writing h_{no} , h_{wrong} and h_{skill} for the state at the last prompt position under no document, a length-matched wrong skill and the gold skill,

$$\underbrace{h_{\text{skill}} - h_{\text{no}}}_{t \text{ (total)}} = \underbrace{h_{\text{wrong}} - h_{\text{no}}}_{g \text{ (presence)}} + \underbrace{h_{\text{skill}} - h_{\text{wrong}}}_{d \text{ (content)}}, \quad (1)$$

an identity. We answer which side does the work by adding each component back to the no-document state and measuring what the model does.

Where to look, and where not to. A patch applied anywhere *other than the document’s own tokens* transports the model’s *decision*, not the document’s content, and how much of a decision a small position group can carry is set by how many tokens the answer occupies (1.000, 0.30 and 0.000 for one token, a short number and a chain of thought). That is a statement about the probe, not the task.

The document’s own span is not subject to that limit, because it is not a small position group. Substituting it—token for token, so every position keeps its rotary phase—recovers 0.87 of the document’s effect on forty clinical calculators at layer 8, against a receiver holding a wrong clinical document that solves none of them. Because a length-matched receiver already holds a document, that substitution satisfies $h_{\text{recv}}(\text{span}) + d(\text{span}) = h_{\text{skill}}(\text{span})$ with $d(\text{span}) = h_{\text{skill}}(\text{span}) - h_{\text{recv}}(\text{span})$, so it is Eq. 1’s content injection at $\alpha = 1$ spread over m positions. Dose, subtraction into shared and specific halves, transfer between documents and truncation of the rank all become available on real skills by running them here.

What is new, and what is not. Several ingredients are established and we use rather than claim them: that a single task vector is functionally a rank-one map (Dong et al., 2026), that a single static position cannot characterise a chain of thought (Dura et al., 2026), that restoring the states of the tokens that carry the information recovers behaviour at early and middle layers and not later (Meng et al., 2022; Geva et al., 2023), that context stops mattering past an intermediate depth (Pola & Balasubramanian, 2025; Sia et al., 2024), and that multiple choice assembles the decision into one position (Wiegrefe et al., 2025). New here is the separation of two limits these leave entangled—the rank of the *mapping* and the length of the *answer*—a channel on which the whole causal battery runs on real documents, and a measurement of the rank of the *representation* that does not scale with width inside a model family and differs between families.

Contributions.

1. **A subtraction protocol with a matched negative control.** Presence dominates the geometry and content the behaviour: presence repairs 0.062 of the rescued cell—exactly what a wrong document in the prompt achieves—while content repairs 0.812 (§3).
2. **Injection capacity is set by answer length** (§3.1): 1.000 for a one-token answer, 0.30 in free-form, 0.000 under a chain of thought, with only the format changed.
3. **A channel on which the whole battery runs on real documents** (§4): over the document’s own span, on forty third-party calculators, $\rho = 0.87$ at $\alpha = 1$ against 0.06 at $\alpha = 0.5$, 0.06 for another skill family, 0.08 permuted, 0.10 for noise, null arm 0.00.
4. **How many directions the effect needs** (§5.2, §5.3): 0.77 at rank 128, nothing for the bottom 64; on a model with a quarter of the width $k^* = 40$ against 52 and a pre-registered width-proportional prediction of 12, but 64–84 on a second family of the same width.
5. **The span stops being a channel rather than a narrow one** (§5.1): writing it at four layers (12–15) recovers 0.96, at twenty (16–35) 0.11, while attention to it still matters far deeper; the dissociation replicates on two tasks and two families. We also retract a mechanism: an apparent collapse of the span’s participation ratio is massive activations (Sun et al., 2024).
6. **Three measurement traps** (§7) and an instrument whose identity intervention changed 6% of chain-of-thought answers until we traced it to a kernel-selection difference and fixed it (§2).

2 SETUP

Material and metric. MedCalc-Bench questions (Khandekar et al., 2024) paired with SRA-Bench skill documents (Su et al., 2026) written by other people—55 calculators $\times$ 20 instances = 1,100 items, graded by the benchmark’s own scorer, answered by a chain of thought. A synthetic positive control (a fictional unit system whose one-page document gives the conversion table) supplies a guaranteed-zero floor and a generatable length-matched control. Causal work is on Qwen3-8B (Yang et al., 2025) unless a model is named; §5.3 adds Qwen3-0.6B, Qwen3-1.7B, Mistral-7B-Instruct-v0.3 (Jiang et al., 2023) and a second domain, TheoremQA from SRA-Bench (747 questions over 320 skills, the benchmark’s prompt and scorer used verbatim). Arms are reported as raw accuracy

and as

$$\rho(\text{arm}) = \frac{\text{acc}(\text{arm}) - \text{acc}(\text{receiver})}{\text{acc}(\text{donor}) - \text{acc}(\text{receiver})}, \quad (2)$$

with intervals bootstrapped over calculators, because a calculator’s twenty instances share a byte-identical document.

Four cells. Every item is labelled by what the document did to it: **R** (rescued), **F** (persistent failure), **K** (kept), **B** (broken). **R is the only cell in which the document demonstrably works;** every table is read by cell, and §7 shows the aggregate over all four can be sign-inverted relative to R. All three arms on the same 1,100 items: gold 74.2%, no skill 34.5%, wrong skill 31.5%.

The null arm, and an instrument fault it found. Every run carries a *null* arm: the receiver’s own span states written back over themselves, an identity that must reproduce the unpatched receiver. For most of this work it did not, quite—on Qwen3-8B, across 1,766 paired observations, it agreed on 93.9% of items, 108 flipping in both directions, with the rate scaling with the model (0.7% on the two smaller ones). The pipeline is deterministic (two machines agreed on all 360 item-by-arm cells), so this was not run-to-run variation. The cause turned out to be an implementation fault rather than a property of the model: the capture pass called the model without an explicit attention mask and the decoding pass with one, which for a single unpadded sequence is mathematically the same but selects a different attention kernel, and the two disagree by 1.2% of the state’s magnitude at layer 8 and 25% at layer 34. Passing the mask in both makes the null arm exact—80/80 items across two reruns. Every span-channel number from §4 on is from runs after the fix unless marked pre-fix; the final-position and synthetic results of §3 and the geometry predate it (Appendix C.3). An identity intervention that is not behaviourally an identity is invisible at one answer token and becomes a six-point floor over a long generation; kernel-dependent numerics are a known source of nondeterminism (He & Thinking Machines Lab, 2025), but we have rarely seen this control reported.

3 WHICH COMPONENT DOES THE WORK, AND WHAT ONE POSITION CAN CARRY

We add each component of Eq. 1 back to the no-document state at the last prompt position and continue the forward pass, against an upper reference that replaces the state outright with h_{skill} . The protocol needs a task with an exact zero floor and a matched negative control, so it is established on the synthetic tier with Qwen3-1.7B (Appendix C.5); this is the only place where the synthetic material carries an argument of its own. **Presence repairs nothing but is not inert**—alone it takes the rescued cell to 0.062, which is exactly what putting the wrong document in the prompt does. **Content repairs most of the cell**, 0.812 against an upper reference of 1.000, having been dismissed by every geometric measurement as a small residual. **And the patch reproduces the document’s whole behavioural signature**: across the four cells, replacement gives 1.000/0.000/1.000/0.000 and adding d_i gives 0.812/0.112/1.000/0.125. A patch that only ever helped would be consistent with a generic “try harder” effect; one that also breaks exactly the items the document breaks is carrying the document’s decision. Which control is subtracted chooses what is measured: against an unrelated document d repairs 0.812, against the same document with every number replaced 0.23–0.31, with its sentences shuffled 0.19–0.29—a factor of three to four, so “the content component” without a named control is not a specified quantity.

3.1 WHAT A SINGLE POSITION CAN CARRY IS THE ANSWER, NOT THE CONTENT

That result is a multiple-choice result, so what does the one position carry? The same items in free-form numeric form collapse: through layer 22 every condition reads zero, including outright replacement, and the best any single-position intervention achieves is 0.304—while the *ordering* is unchanged, content 0.268 of that 0.304 against presence 0.018. It is the format, not the model: the identical sweep on the *same model and items* in multiple-choice form recovers 1.000. The reason can be counted. Of the 120 items none has a single-token gold answer (36 are two tokens, 55 three, 29 four), and a patch at one prompt position can fix at most the first generated token; multiple choice compresses the answer to exactly one token, which is the unit a single-position patch

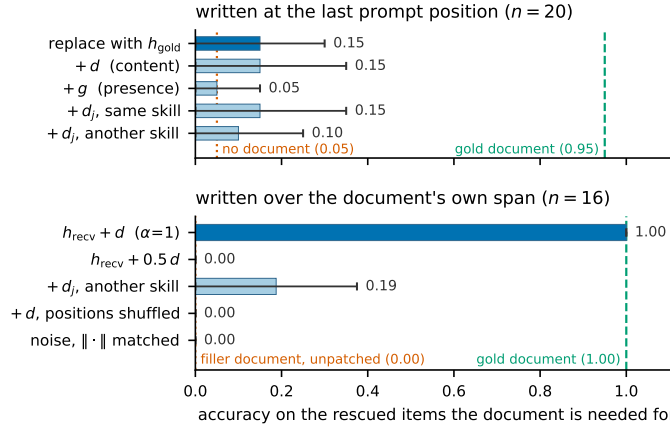


Figure 1: The same content vector, in two places. Third-party clinical skills, Qwen3-8B, answers graded from the model’s own reasoning. Top: written into the last prompt position, each arm at the best of layers 10, 14, 20, 30, against a no-document floor. Bottom: written over the document’s own span at layer 8, against the floor its own receiver sets. The bottom panel is restricted to the calculators whose receiver—a wrong clinical document—never solves an item, because on the others the model does not need the document and a coherent replacement lets it fall back on what it already knows; the unrestricted version and that comparison are in Appendix D. Error bars are bootstrap 95% intervals.

transports. Patching the last k positions confirms a capacity rather than a threshold: $k = 1, 4, 16$ recover 0.286, 0.286, 0.554.

Making answer length the *only* thing that moves settles it. The same 80 items, same model, same document, prompted to reason step by step: the skill is worth $0.000 \rightarrow 1.000$, so there is more effect to recover than in either other format, and the single-position patch recovers **0.000**. Over the three formats the rescued cells hold 44, 56 and 80 items and the patch recovers 1.000, 0.304 and 0.000. On third-party clinical skills the same probe reads nothing—replacing the final position repairs at most 3 of 18 rescued items (layer 10) and 0–1 from layer 14 on, and transplanting the *question* span, 50 to 1697 positions into a length-matched receiver, repairs nothing at any of seven layers, with the null arm on the receiver’s baseline to the item.

4 THE CHANNEL THAT DOES CARRY, AND THE WHOLE BATTERY INSIDE IT

The receiver, and why the transplant is an injection. The donor is the with-document forward; the receiver is the same prompt with the document replaced, token for token, by a fixed unrelated clinical skill, so every position keeps its rotary phase and the receiver is a live distractor rather than filler. Writing the donor’s states over those positions therefore satisfies $h_{\text{recv}} + d = h_{\text{skill}}$ on the span: it is d injected at $\alpha = 1$ over m positions. Every arm below is a vector operation on $d(\text{span})$. It is not a re-run of the gold prompt: it is applied at one layer, everything outside the span keeps the receiver’s states, and it recovers 0.85 at layer 8 and 0.19 at 16 (§5.1).

It carries, completely. Figure 1 puts the two channels side by side on the same items. At the final position five arms—the content component, the presence component, a donor from another instance, one from another skill, and outright replacement—all land within 0.10 of a model given no document. Over the document’s own span the transplant takes the items that need the document from 0.00 to 1.00 at layer 8 ($n = 16$; 0.87 on forty calculators, Table 1). The difference is not the vector; it is where it is written.

It behaves like a mechanism, not a perturbation. Four controls, all at layer 8, all read on the calculators a wrong document never rescues. *Dose* is a step, not a ramp: $\alpha = 0.4, 0.5, 0.55$ give 0.06, 0.00, 0.06 and 0.6, 0.7, 1.0 give 0.62, 0.94, 1.00 ($n = 16$, four calculators)—the whole transition lies between 0.55 and 0.7 of a dose. *Transfer*: written over the prefix two documents share, an-

Table 1: The span battery on forty calculators. The restricted split—the fifteen calculators a wrong document never rescues—is recomputed inside this set, so no number is inherited from the ten-calculator version. Intervals are bootstrapped over calculators. The last two rows are the instrument: writing the receiver’s own states back into it, and not patching at all.

donor written over the document’s span	accuracy	CI ₉₅	fraction
<i>the gold document in context</i>	0.867	[0.781, 0.953]	+1.00
$h_{\text{recv}} + d$ ($=h_{\text{skill}}, \alpha = 1$)	0.750	[0.640, 0.860]	+0.87
the same d , only over the prefix shared with another skill	0.633	[0.511, 0.755]	+0.73
$h_{\text{recv}} + \frac{1}{2}d$	0.050	[0.000, 0.105]	+0.06
$\bar{d}$, a skill in another family	0.050	[0.000, 0.105]	+0.06
$\bar{d}$, a skill in the same family	0.050	[0.000, 0.146]	+0.06
the same d , span positions permuted	0.067	[0.004, 0.130]	+0.08
noise, $\ \cdot\ $ matched per position	0.083	[0.013, 0.153]	+0.10
h_{recv} itself (the null arm)	0.000	[0.000, 0.000]	+0.00
<i>the filler document, unpatched</i>	0.000	[0.000, 0.000]	+0.00

other skill’s vector gives 0.19 against 0.94 for this skill’s own vector on the same positions ($n = 16$), and on forty calculators 0.06 against 0.73 whether or not the donor computes the same quantity, so the failure is the donor, not the sub-span. *Order*: the same vectors permuted across the span’s own positions give 0.08. *Noise*: norm-matched Gaussian gives 0.10, against a null arm at 0.00 (both on the forty calculators of Table 1).

The split these are read on. On six of the ten development calculators the receiver—a wrong clinical document—already solves some item: the model knows that calculator and only needed telling which one to run. There the document is not needed, and a donor from another skill reads $\rho = 0.47$ against 0.19 on the rest ($n = 24$ and 16), so a transfer arm read over the mixture reports as transfer what is partly *release*. Every causal number is read on the calculators whose receiver never solves an item—a condition on a control, never on the donor or the treatment.

It replicates on four times as many skills. Run on *forty* calculators with the restriction recomputed inside the larger set, fifteen depend on the document, leaving 60 items, and every arm holds (Table 1).

The span is 415 to 1,630 positions wide on the forty calculators, the same order as the answer it has to support, which with §3.1 gives the rule the paper is built on: *a patch transports what fits in the positions it is given*.

5 WHERE IN THE NETWORK THE CONTENT IS READABLE

5.1 SUFFICIENCY AND NECESSITY DISAGREE ABOUT DEPTH

Can the document’s states be handed over at layer L , and *is* the model still reading them at L , give different answers, on every model and task we ran.

Sufficiency, on forty calculators ($n = 60$ after the restriction, twenty calculators): a single layer’s worth of the span reproduces $\rho = 1.04, 0.94, 0.85, 0.77$ at layers 0, 4, 8, 12 of 36, then falls—0.28 at 14, 0.19 at 15 and 16, 0.15 at 20. The handover is finished by about 0.4 of the way in.

Necessity, from cumulative attention knockout ($n = 40$): masking every position’s attention to the document from layer L to the end leaves 0.08–0.15 of the items correct for every $L \leq 20$ but $L = 4$ (0.33), then 0.38 at 24, 0.75 at 28 and 0.95 at 32. **The model is still reading the document at 0.7 of the way in, long after a single-layer handover has stopped working at 0.4.** The dissociation replicates on every model and task we ran (§5.3, Table 2); where the boundaries sit does not. Within the Qwen family the handover ends at the same relative depth on two tasks and two widths (0.33–0.43); Mistral-7B hands over far earlier (0.19–0.22 on both tasks) yet is still reading the document at 0.62–0.88; and the necessity boundary moves with the task (0.56 on TheoremQA against 0.78 on MedCalc for Qwen3-8B).

It is not that one layer is too few: the channel closes. Writing the span at *every* layer at once recovers 0.950 (pre-fix), which makes “one layer is not enough” the natural reading. It is wrong. Writing the span at every layer in a window and nowhere else, on twenty calculators ($n = 48$), gives $\rho = 1.00$ for layers 8–11, 0.96 for 12–15, 0.38 for 14–19, which straddles the boundary, and 0.11 for 16–35—the same as a single layer at 16, and against a null arm at exactly 0.00. **Four layers are enough if they are 12 to 15; twenty are worth nothing if they start at 16.** On TheoremQA the same windows give 0.90, 0.73, 0.45 and 0.21, the last equal to layer 16 alone. Past the handover those positions are not a route into the computation—not a narrow one, no route.

The span does *not* become a summary; the measurement that says it does is an artefact. The centred participation ratio of the span’s content matrix reads 64 at layer 12 and 17 at layer 16, as if the document were compressed exactly where the handover dies. It is a step at one layer, absent in Qwen3-0.6B and Qwen3-1.7B, and it is massive activations (Sun et al., 2024; Shi et al., 2026): at layer 16 one token in 34 of 40 documents acquires a state 99 to 180 times the span’s median norm on the same two hidden dimensions. Removing the four loudest of four hundred positions restores the ratio to 66, and measured that way it is flat from layer 0 to 28 (Appendix Table 6). The handover dies without the representation losing dimensionality, and we do not know what it does lose. Any spectral statistic of a residual stream that does not exclude massive activations (Fernando & Guitchoints, 2026; Sarkar & Deka, 2026) reports a collapse at the emergence layer, as rogue dimensions do for similarity (Timkey & van Schijndel, 2021; Kovaleva et al., 2021); we recommend reporting the per-position norm ratio beside it (here $1.6 \rightarrow 150$, medians).

5.2 HOW MANY DIRECTIONS THE EFFECT NEEDS: WE TRUNCATE THE RANK

Every measurement so far says *where* the effect lives. This one says *how big* it is, in the only unit comparable across the task-vector literature: how many directions of the injected object the model uses. With $d \in \mathbb{R}^{m \times D}$ and $\bar{d}$ its mean over positions,

$$d^{(k)} = \bar{d} + \sum_{i=1}^k \sigma_i u_i v_i^\top, \quad d - \bar{d} = \sum_i \sigma_i u_i v_i^\top, \quad (3)$$

the mean kept and only the variation truncated, so k is comparable with a centred participation ratio. Note what the retained mean means: every truncated arm already contains *one vector for this document written at every position of its span*, which is the most generous form a task vector could take here. The bottom-64 control, whose directions carry almost none of the energy, is effectively that mean alone.

One direction is not enough and neither are sixteen (Figure 2). At layer 8 on forty calculators ($n = 60$) recovery is 0.13, 0.06, 0.13 at $k = 4, 16, 32$, then 0.54 at 64 and 0.77 at 128 against 0.87 untruncated; half of the curve’s range above $k = 16$ is recovered at $k^* = 52$ [43, 72], 1.3% of the width. The ten-calculator development curve, run before the fix, adds $k = 1, 2, 8$ (0.11, 0.06, 0.11) and puts k^* at 45. Keeping the *bottom* k directions instead—the same rank, the discarded half of the spectrum—gives 0.06 at $k = 16$ and 0.00 at $k = 64$: it is the top of the spectrum, not the count. Repeated deeper, at layer 12 and, with the document after the question, at 16, both knees sit between 64 and 128, not lower (pre-fix, $n = 16$ and 28; Appendix C.1).

5.3 A NARROWER MODEL, A SECOND FAMILY, A SECOND TASK

We repeated the channel, the depth sweep, the knockout and the truncation on Qwen3-0.6B ($d = 1024$) and Mistral-7B (32 layers, $d = 4096$), and all of them on TheoremQA, injecting before each model’s own handover and recomputing the restricted split per model and task—“the receiver never solves this group” is a statement about a model (Qwen3-1.7B, whose document rescues only 8–11% of items, is in Appendix Table 4). TheoremQA skills mostly serve one question, so there the restriction is per item, which the fixed instrument permits: the receiver is deterministic and the null arm reproduces it exactly.

Three things replicate everywhere (Table 2). *The channel*: ρ between 0.74 and 0.94 in all six. *Its specificity*: every control that ran stays at or below 0.27, no more than a third of the transplant; the floor is higher on TheoremQA, where the restriction is per item, and on Mistral-7B than on

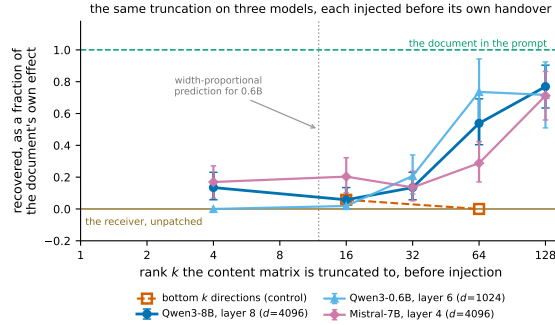


Figure 2: Truncating the content matrix to rank k before injecting it, on forty calculators after the instrument fix, each model injected before its own handover. Qwen3-0.6B has a quarter of Qwen3-8B’s width and its knee is not lower ($k^* = 40$ against 52; a width-proportional account puts it at the dotted line). Mistral-7B has Qwen3-8B’s width and needs more ($k^* = 84$). The dashed control keeps the *bottom* k directions—the same rank, the discarded half of the spectrum. ρ is normalised by the document in the prompt; the untruncated transplants read 0.87, 0.89 and 0.92. Intervals are bootstrapped over calculators.

Table 2: Every core measurement, per task and model, after the instrument fix. ρ : the span transplant at the injection layer (Qwen3-8B 8, Qwen3-0.6B 6, Mistral-7B 4). Max control: the largest of half dose, another skill, permuted positions and noise. Handover: relative depth between the last layer with $\rho \geq 0.5$ and the first below. Reading: relative depth of the first knockout layer that keeps at least half the rescued items. k^* : half of the rank curve’s range above $k = 16$. “—”: not run after the fix.

task	model	n (groups)	ρ [95% CI]	max control	handover	reading	k^*
MedCalc	Qwen3-8B	60 (15)	0.87 [0.73, 0.98]	0.10	0.33–0.39	0.78	52
TheoremQA	Qwen3-8B	99 (75)	0.84 [0.75, 0.92]	0.25	0.33–0.39	0.56	50
MedCalc	Qwen3-0.6B	204 (35)	0.89 [0.70, 1.09]	—	—	—	40
TheoremQA	Qwen3-0.6B	132 (97)	0.74 [0.66, 0.82]	—	0.36–0.43	—	52
MedCalc	Mistral-7B	92 (26)	0.92 [0.77, 1.09]	0.27	0.19–0.22	0.88	84
TheoremQA	Mistral-7B	115 (94)	0.94 [0.84, 1.05]	0.27	0.19–0.22	0.62	64

MedCalc with Qwen3-8B. *The dissociation*: wherever both sweeps ran, the handover ends long before attention to the document stops mattering, most widely on Mistral-7B.

What does not replicate is where the numbers sit. The handover’s relative depth is shared within the Qwen family and not across families (§5.1). And the number of directions: the prediction registered before the Qwen3-0.6B run was that either k^* is constant across widths or $k^* \propto d$, which puts the 0.6B knee at $k \approx 12$. After the fix it reads $\rho = 0.02, 0.21, 0.74, 0.72$ at $k = 16, 32, 64, 128$, so $k^* = 40$ [32, 50] against Qwen3-8B’s 52 [43, 72]; on TheoremQA, 52 against 50. Width-proportional scaling is rejected on both tasks, and the content matrix’s participation ratio agrees (66–77 on Qwen3-0.6B, 70–99 on Qwen3-8B over layers 0–20, massive activations excluded). But Mistral-7B, at Qwen3-8B’s width, needs 84 [59, 97] on MedCalc and 64 [50, 79] on TheoremQA—higher on both, clearly so on the first—and its $k = 128$ arm is still well short of its untruncated transplant (0.71 against 0.92), so our grid bounds its knee. The number of directions a document needs does not scale with width inside a family; it is not a property of the document alone. Inside the family the cost is not constant: the same document occupies about 4% of Qwen3-0.6B’s residual width and 1.3% of Qwen3-8B’s.

6 WHAT THE CONTENT COMPONENT BELONGS TO

Four properties, each an arm in the same battery, each on the restricted split; Appendix C.13 gives the numbers and the controls. It is **not a common “a document is present” shift**—averaging d over every skill and writing that average recovers $\rho = 0.11$ (0.06 with the document last; pre-fix) while

accounting for 54–60% of each vector’s energy. It is **not a magnitude**—rescaling per position to the receiver’s own norm changes nothing (1.06 against 1.06; pre-fix). It is **the skill and not the skill-plus-task**: with the document ahead of the question causal masking makes the span’s states identical across a skill’s twenty instances (bit for bit at equal prompt length, cosine ≥ 0.997 otherwise), so the vector cannot contain task information because it cannot see the task; placed *after* the question, where it can, another instance’s vector still recovers as much as the instance’s own at layer 8 (0.96 against 1.00, $n = 28$) and the instance-specific remainder recovers 0.04. That holds at layer 8 only: at 16 another instance’s vector recovers 0.48 against the instance’s own 0.85 while the remainder alone still recovers 0.04, so by then the span has taken up something of the question that neither part carries by itself. And it is **concentrated in the formulas**. Decoding the token split names the four quarters: a prose description, the formulas, the tool signatures, a worked example. Written one at a time on forty calculators ($n = 60$) they recover 0.12, **0.56**, 0.12 and 0.12: three quarters are each at the control floor, yet the formulas alone reach only two thirds of the whole document’s 0.87, so the rest contributes only in combination. Moving the document behind the question keeps the handover open at layer 16 (0.85 on ten calculators, against 0.19 on forty with it first), so the handover depth is not a fixed property of the network.

7 THREE WAYS THIS MEASUREMENT MISLEADS

Three of our measurements were wrong before they were right (Appendix C.8). **An aggregate log-probability ranks the controls first**: averaged over all items, at the last layer, the gold answer’s log-probability puts a content-free mean vector 3.75 nats *above* the state the model actually builds while reading the document, with accuracy 0.256 against 0.436—a flattened distribution never assigns −19 nats and never reaches the argmax; at layer 21 the cells where the document does not work contribute 133% of the gap. **Multiple choice manufactures a floor**: a document with every number replaced keeps nearly three quarters of the effect there and exactly none in free-form, because the format scores retrieval as success (Balepur et al., 2024). **The wrong-document control is not the no-document control**: any document in the slot moves rescued items off the floor (0.025 with none, 0.150–0.250 with an irrelevant one), so “with” against “without” measures presence and content together—which is why every arm here is measured against a receiver holding a document.

8 RELATED WORK

What one vector can carry. That a context compresses into one activation-space direction is established for in-context demonstrations (Hendel et al., 2023; Todd et al., 2024) and, in weight space, for fine-tuning (Ilharco et al., 2023). Two limits on that move are known and they are not the same limit. Dong et al. (2026) prove that one injected task vector is functionally a rank-one map, which *explains* our finding that one vector shared by a whole item set recovers 0.062 on a 39-row lookup. The limit this paper is about is on the other side of the map: the item’s own d_i recovers 0.812 through the same single position *despite* that same high-rank mapping, because the answer is one token, and nothing on clinical skills because the answer is hundreds. Rank constrains which mappings one vector can express; length constrains how much of a decision any small position group can carry, whatever the mapping (Dura et al., 2026 assert the latter; §3.1 measures it). The strongest positive result for single-vector compression, Ardoin et al. (2026), is convergent on both counts: every task it succeeds on has a short answer, and its optimal configuration injects at an *early* layer. With Wiegrefe et al. (2025), who localise the choice of an answer *symbol* to one middle layer, the three say a single vector is a *selector* among behaviours the model already has. Trained compressors do carry long contexts through a few positions (Mu et al., 2023; Ge et al., 2024), and a direction added at *every* position steers long generations (Panickssery et al., 2024; Liu et al., 2024; Stolfo et al., 2025); our limit is on an untrained write at one position, and that this position *encodes* properties of the coming response (Dong et al., 2025) is a decoding result, not a transport one. Mehrafarin et al. (2026) show what the answer position does hold—one late position patched from the model’s *own completed* chain of thought recovers GSM8K at $\geq 99\%$, while our donor, taken before any reasoning, recovers nothing—and Da Silva et al. (2025) report the same channel failing from another direction.

Four things are called “rank”; and patching. Dong et al. (2026) bound the rank of the *task map*; Sarkar & Deka (2026) that of the *Jacobian*, reporting it layer-intrinsic and unchanged with model size; Fernando & Guitchounts (2026) that of the *composed operator*, finding a $65\times$ collapse. We measure the rank of a *representation*, $d \in \mathbb{R}^{m \times d}$, and are the only one of the four to truncate the injected object itself and read out behaviour. A caution applies to the descriptive ones, ours included: Sun et al. (2024) document activations three orders of magnitude above the median at fixed dimensions and a few positions, appearing at a single layer (Shi et al., 2026), so a spectral statistic on a residual stream past that layer is a statistic about them (§5.1). On method we follow Zhang & Nanda (2024) and Heimersheim & Nanda (2024); Pola & Balasubramanian (2025) and Sia et al. (2024) find a layer past which the context stops mattering, which is our knockout result from the opposite direction and what the shallow-read, deep-extract flow of Wang et al. (2023) predicts; and Li et al. (2025), Abdelsalam et al. (2026) and van Dort & Heuss (2026) localise transferable task signal without separating presence from content. A parallel line replaces the document with weights rather than activations (Charakorn et al., 2026; Caccia et al., 2025).

9 LIMITATIONS

Scope. The battery was developed at $n = 40$ and rerun on forty calculators, a second domain and two further models; the conclusions are the ones that survived. The channel, depth sweep and truncation have three models and two tasks, the knockouts two models, the document-order comparison and the localisation one model and one task; the width argument rests on a factor of four within one family, and k^* on a grid that bounds Mistral-7B’s knee. Two further SRA-Bench domains mark where the method stops: on LogicBench’s mostly binary answers ($n = 57$) half a dose recovers 0.44 and another skill 0.38 against the transplant’s 0.75, with no cliff—a perturbation flips a two-way answer by chance—and CHAMP leaves 20 rescued items. ToolQA and BigCodeBench need tools or sandboxed execution and were excluded. Results predating the instrument fix—the synthetic tier, the final-position sweeps, the ten-calculator rank curve and its deeper repeats—are marked in Appendix C.3.

Which items the causal claims are about. The rescued cell, and within it the groups whose receiver never solves an item (§4): fifteen calculators and 60 items of forty. Both conditions are on controls.

What the rank and window results do not establish. We do not claim that what a deep layer lacks is rank, and a truncated patch can open pathways the unpatched model does not use (Makelov et al., 2024). That writing the span at layers 16–35 restores nothing is also compatible with late reading that depends on bindings formed early (Feng & Steinhardt, 2024); we have not separated the two. Two negative results: no representational quantity at the final position distinguishes the items a skill rescues from those it does not, and the span channel has no usable log-probability reading (a forced ANSWER: cue removes the behavioural effect entirely while leaving a +0.60 nat difference), so everything here is on accuracy.

10 CONCLUSION

A skill document’s effect is not carried by a vector at the position the model answers from. That position holds the model’s *decision*, and a decision occupying hundreds of tokens does not fit into one state—which is why every arm written there, replacement included, lands on the no-document baseline, and why the effect reappears the moment the answer is made short. It is carried by the document’s own tokens, where the same subtraction behaves like a mechanism: complete at $\alpha = 1$, switching on between 0.55 and 0.7 of a dose, indifferent to the norm written, destroyed by permuting the span’s positions, absent for a donor from another skill, on two task domains and two model families, and requiring fifty to eighty of the injected object’s directions—a number that does not scale with width inside a family but differs between families. The practical reading is a scope condition: activation patching at the final prompt position measures what the model has decided, and on short-answer benchmarks that is indistinguishable from measuring what the document supplied.

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Table 3: The same decomposition against three control documents. Synthetic control, multiple choice, Qwen3-1.7B, $n = 200$, rescued cell $n_R = 48$, layers 21–27. “in context” is the control document actually placed in the prompt; “+ g ” is its state injected at the final position of the no-document prompt. That the two agree to within 0.02 for three very different documents is the strongest validity check we have on the patch itself. d is the same measurement in all three rows; only what it is subtracted against changes.

control document	fraction of the rescued cell correct		
	control in context	$h_{\text{no}} + g$	$h_{\text{no}} + d$
unrelated text, matched length	0.062	0.062	0.812
same document, lines scrambled	0.625	0.604–0.625	0.188–0.292
same document, factors replaced	0.729	0.708–0.750	0.229–0.312
gold skill in context	1.000	—	—
no document	0.000	—	—

A CONTROL DOCUMENTS IN FULL

d is defined by the document it is subtracted against, so “content” is whatever separates the skill from that control. Our default control holds only length fixed. Two tighter ones, generated from the skill document itself, hold more: **shuffled** keeps every word and every number and destroys the line order; **corrupted** keeps the wording, the procedure, the table layout and the ordering of the units, and replaces *every conversion factor*. All three are injected under the same name and header, so the prompt never announces which it is.

Both tighter controls move the accounting a long way (Table 3). A table whose numbers are all wrong solves 0.729 of the rescued cell; a table with the right numbers in scrambled order solves 0.625; the unrelated text solves 0.062. Against no document the cell is 0.000 by construction and with the gold document it is 1.000. So when either tight control is used, g is no longer inert—it repairs 0.60–0.75— and the residual d repairs only 0.19–0.31.

The same control on third-party skills. We repeated the corrupted control on the clinical calculators, where the arm already exists in the released harness (numeric constants in the prose perturbed, wording, structure and executable tools untouched). That material is free-form and answered through a chain of thought, and there the corrupted skill is worth *nothing*: on 20 rescued items it scores 0.050, which is what no skill at all scores, against 0.950 for the gold skill. The synthetic and the real material therefore agree, and they agree on the thing that matters—whether corrupting a document costs you anything is a property of the answer format, not of the document.

Read as an attribution rather than as three competing estimates, this says something uncomfortable about the multiple-choice form of the task. On the full 200 items the gold document is worth $0.180 \rightarrow 0.300$; a scrambled version of it is worth $0.180 \rightarrow 0.270$ and a version with every factor wrong is worth $0.180 \rightarrow 0.260$. Three quarters of the measured skill effect survives destroying either the order of the document or the correctness of every number in it. What the model needs, in this format, is mostly a document containing the right vocabulary and the right kinds of number—not a correct table. The correct table is worth the remaining 0.03–0.04 absolute, or 0.19–0.31 of the rescued cell.

That is a statement about the answer format as much as about the model. We registered the test before running it: in the free-form format the number has to be produced, so a corrupted table should be worth nothing. It is (Table 10). Corrupting every factor keeps 0.67 of the multiple-choice effect on the full item set and *none* of the free-form effect’s 0.467. Scrambling the line order costs little in either, which is what it should: our shuffle permutes lines, and each row of the conversion table is one line, so every factor is still there to be retrieved—just not in a readable order.

B THE GEOMETRY AT THE FINAL POSITION, IN FULL

Table 7 says the effect is content: a document that is present but wrong is worth nothing. The internal picture at the last prompt position says the opposite, and it says so exactly where the document is being read. The cosine between the displacement caused by the skill and the displacement caused by an unrelated document of similar length is 0.93 at layer 0, 0.9475 at layer 4 and 0.91 at layer 10 on the synthetic control, falling to 0.35–0.41 only from layer 18, i.e. after the model has stopped reading (§5). On a second family the same comparison at layer 10 gives 0.9716 between two mutually exclusive skills against 0.9445/0.9555 between each of them and an unrelated document—a gap of 0.016–0.027 with bootstrap intervals of width 0.001, so it is real and it is negligible.

Read alone, that says the injection is not content-specific. Under Eq. 1 it says something narrower and correct: over the first half of the network $\|d\|$ is only 0.38–0.45 of $\|t\|$, so most of what a cosine between h_{skill} and h_{wrong} measures is shared, and the interesting quantity is the part that is not. Three facts follow, all in Figure 8 with the numbers in Appendix D. The content component is a minority of the displacement wherever the document is being read—under half of $\|t\|$ through layer 14. Through layer 16 $\|g\|$ actually exceeds $\|t\|$: the *wrong* document moves the state further than the right one, and the two components partly cancel, which is why the ratio recovers deeper without g shrinking. And two items’ content vectors are far more aligned when they use the same row of the conversion table than when they do not, so whatever d encodes is about the document’s content rather than about the item.

None of this is causal evidence, and the rest of the paper is about why it should not be read as any. A displacement can be large and do nothing; a direction can differ systematically between documents and carry none of its effect (§6). We use the geometry only to say which quantity is worth intervening on, and then intervene.

C FIGURES AND TABLES MOVED FROM THE MAIN TEXT

Table 4: The channel and the truncation curve at three model widths, each with its own restricted split and its own injection layer at matched relative depth. ρ is Eq. 2; *gold* is what the document itself is worth on that model’s restricted items, which is why the 0.6B and 1.7B rows have low absolute accuracy and a large n : those models solve almost nothing without the document, so almost every item qualifies. Qwen3-0.6B and Qwen3-8B are after the instrument fix; Qwen3-1.7B is before it, and its curve is not monotone, so its k^* is not interpretable. k^* sits between 32 and 64 at $d = 1024$ and at $d = 4096$ alike; a width-proportional account predicts $k \approx 12$ for the first. Mistral-7B is in Table 5.

model	d	layers	inject	n	gold	$\rho(\text{full})$	$k=16$	32	64	128	bottom-64	k^*
Qwen3-0.6B	1024	28	6	204	0.260	+0.89	+0.02	+0.21	+0.74	+0.72	+0.02	40
Qwen3-1.7B	2048	28	6	267	0.090	+0.54	+0.33	+0.50	+0.42	+0.29	+0.12	23
Qwen3-8B	4096	36	8	60	0.867	+0.87	+0.06	+0.13	+0.54	+0.77	+0.00	52

C.1 RANK TRUNCATION AND THE MODEL LADDER, IN MORE DETAIL

Every measurement so far says *where* the document’s effect lives. This one says *how big it is*, in the only unit that is comparable across the task-vector literature: the number of directions of the injected object that the model actually uses.

We truncate the content matrix across the span’s positions to rank k before writing it. With $d \in \mathbb{R}^{m \times D}$ and $\bar{d}$ its mean over positions,

$$d^{(k)} = \bar{d} + \sum_{i=1}^k \sigma_i u_i v_i^\top, \quad d - \bar{d} = \sum_i \sigma_i u_i v_i^\top, \quad (4)$$

the mean kept and only the variation truncated, so that k is comparable with a centred participation ratio. $k = 1$ is the closest thing in this setting to a task vector: one direction, applied at every position of the span.

One direction is not enough, and neither are sixteen (Figure 2). On forty calculators after the fix ($n = 60$), recovery at layer 8 is 0.13, 0.06, 0.13, 0.54, 0.77 at $k = 4, 16, 32, 64, 128$ against 0.87 untruncated, and half of the curve’s range above $k = 16$ is recovered at $k^* = 52$ (95% bootstrap interval over calculators [43, 72]; interpolated in $\log_2 k$, the definition fixed before the model ladder ran). The ten-calculator development curve, before the fix, reads 0.11, 0.06, 0.06, 0.11, 0.11, 0.33, 0.72, 0.94 at $k = 1, 2, 4, 8, 16, 32, 64, 128$ and gives $k^* = 45$.

k^* is anchored to the largest value on the grid $k \leq 128$, not to the untruncated transplant. Where $k = 128$ is close to untruncated (Qwen3-8B, 0.77 against 0.87) that makes little difference; where it is not (Mistral-7B, 0.71 against 0.92) the knee is bounded by the grid, and a finer or longer grid could move it up.

It is the top of the spectrum, not the count. Keeping the *bottom* k singular directions instead—the same rank, the discarded half of the spectrum—gives $\rho = 0.06$ at $k = 16$ and 0.00 at $k = 64$ on forty calculators (0.00 and 0.06 on the development set), both at the receiver. Without that control, “sixty-four directions are enough” could not be distinguished from “any sixty-four directions will do”.

The requirement does not fall with depth (pre-fix, small n). If the handover died at layer 15 because the object had shrunk, the number of directions needed would shrink with it. Repeating the truncation where the transplant still works, on the calculators the receiver never solves, gives accuracy 0.00 at $k = 16$ and 0.62 at $k = 128$ against 1.00 untruncated at layer 12 ($n = 16$), and 0.25 at $k = 16$ and 0.71 at $k = 128$ against 0.79 at layer 16 with the document after the question ($n = 28$)—the knee between 64 and 128 in both. These runs predate the fix and are small; we read them as “not lower”, not as a measured knee. At layer 16 with the document *before* the question, where the transplant is already at the floor ($\rho = 0.06$), there is no curve at all: every truncation reads

between 0.00 and 0.06, because there is no longer an effect for truncation to remove. We do not read the rank of the layer-16 doc-first object at all, since that is the layer at which the massive activations of §5.1 take over its spectrum.

The obvious converse does not work, and we report that. If a deep layer’s handover failed for want of directions, supplying it with a higher-rank message should restore it. We tried the direct version—write the layer-8 donor into layer 16—on ten calculators ($n = 30$ after restricting to items that neither the wrong document nor the no-document prompt solves). It does not restore; it reads $\rho = 0.00$ where the same-layer donor reads 0.06, and matching the norm to layer 16’s scale recovers only part of the gap (0.13). The layer-8 state is not a high-rank version of the layer-16 state, it is a different code, and the transfer is directional: Ardoin et al. (2026) find middle-to-early injection optimal, and we find early-to-middle injection harmful.

C.2 REPLICATION ACROSS TASKS AND MODEL FAMILIES, IN FULL

Every point behind Table 2. All runs are after the instrument fix. MedCalc rows use forty calculators and each model’s own behavioural screen; TheoremQA rows use SRA-Bench’s 747 questions with its prompt and scorer verbatim, and since most of its 320 skills serve a single question the restriction there is per item. Knockout rows are fractions of each model’s rescued items kept when attention to the document is blocked from layer L to the end, unrestricted.

Table 5: The replication curves. Battery, sufficiency and rank are ρ (Eq. 2) on the restricted split; necessity is the fraction of rescued items kept. “Another family” (MedCalc) and “another skill” (TheoremQA) are the transfer controls available in each set.

task / model	measurement	values
MedCalc / Qwen3-8B	battery, L8 ($n = 60, 15$ groups)	transplant 0.87, own d on shared prefix 0.73, $\alpha=0.5$ 0.06, another family 0.06, permuted 0.08, noise 0.10, null 0.00
	sufficiency ($n = 60$)	L0 1.04, L4 0.94, L8 0.85, L12 0.77, L14 0.28, L15 0.19, L16 0.19, L20 0.15
	necessity ($n = 40$ rescued)	L0 0.12, L4 0.33, L8 0.15, L12 0.10, L16 0.07, L20 0.07, L24 0.38, L28 0.75, L32 0.95
	rank, L8 ($n = 60$)	$k=4$ 0.13, $k=16$ 0.06, $k=32$ 0.13, $k=64$ 0.54, $k=128$ 0.77, full 0.87, bottom-16 0.06, bottom-64 0.00, k^* 52
TheoremQA / Qwen3-8B	battery, L8 ($n = 99, 75$ groups)	transplant 0.84, own d on shared prefix 0.86, $\alpha=0.5$ 0.19, another skill 0.25, permuted 0.24, noise 0.22, null 0.00
	sufficiency ($n = 99$)	L0 1.00, L4 0.94, L8 0.84, L12 0.61, L14 0.37, L16 0.19, L20 0.14
	necessity ($n = 106$ rescued)	L0 0.31, L4 0.32, L8 0.38, L12 0.35, L16 0.38, L20 0.53, L24 0.70, L28 0.83, L32 0.90
	rank, L8 ($n = 99$)	$k=4$ 0.17, $k=16$ 0.24, $k=32$ 0.30, $k=64$ 0.54, $k=128$ 0.67, full 0.84, bottom-16 0.10, bottom-64 0.13, k^* 50
MedCalc / Qwen3-0.6B	rank, L6 ($n = 204$)	$k=4$ 0.00, $k=16$ 0.02, $k=32$ 0.21, $k=64$ 0.74, $k=128$ 0.72, full 0.89, bottom-16 0.02, bottom-64 0.02, k^* 40
TheoremQA / Qwen3-0.6B	sufficiency ($n = 132$)	L0 0.97, L3 0.88, L6 0.74, L8 0.69, L9 0.65, L10 0.64, L12 0.33, L16 0.28
	rank, L6 ($n = 132$)	$k=4$ 0.06, $k=16$ 0.09, $k=32$ 0.14, $k=64$ 0.43, $k=128$ 0.61, full 0.74, bottom-16 0.06, bottom-64 0.08, k^* 52
MedCalc / Mistral-7B	battery, L4 ($n = 92, 26$ groups)	transplant 0.92, own d on shared prefix 0.69, $\alpha=0.5$ 0.27, another family 0.24, permuted 0.27, noise 0.15, null 0.00
	sufficiency ($n = 79$)	L0 1.00, L1 0.98, L2 1.00, L3 1.02, L4 0.98, L5 0.90, L6 0.76, L7 0.49, L10 0.35, L12 0.22, L14 0.24, L16 0.22, L20 0.10, L24 0.06
	necessity ($n = 72$ rescued)	L0 0.11, L4 0.17, L8 0.12, L12 0.24, L16 0.24, L20 0.26, L24 0.42, L28 0.58
	rank, L4 ($n = 92$)	$k=4$ 0.17, $k=16$ 0.20, $k=32$ 0.14, $k=64$ 0.29, $k=128$ 0.71, full 0.92, bottom-16 0.08, bottom-64 0.17, k^* 84
TheoremQA / Mistral-7B	battery, L4 ($n = 115, 94$ groups)	transplant 0.94, own d on shared prefix 0.80, $\alpha=0.5$ 0.27, another skill 0.20, permuted 0.17, noise 0.13, null 0.00
	sufficiency ($n = 115$)	L0 1.01, L2 0.96, L4 0.94, L5 1.00, L6 0.82, L7 0.49, L10 0.30, L14 0.11, L20 0.05
	necessity ($n = 82$ rescued)	L0 0.09, L4 0.10, L8 0.10, L12 0.26, L16 0.48, L20 0.72, L24 0.85, L28 0.89
	rank, L4 ($n = 115$)	$k=4$ 0.17, $k=16$ 0.20, $k=32$ 0.23, $k=64$ 0.48, $k=128$ 0.76, full 0.94, bottom-16 0.14, bottom-64 0.13, k^* 64

Three observations are not in the main text. (i) On TheoremQA with Qwen3-8B, blocking attention to the document from layer 0 still keeps 0.31 of the rescued items, against 0.12 on MedCalc. Blocking attention is not removing the document—its positions still shift the question’s rotary phase and length—and the per-item restriction selects items near the model’s decision boundary; we have not

separated the two. (ii) The same per-item selection is the most likely reason TheoremQA’s controls sit at 0.19–0.25 for Qwen3-8B. (iii) On LogicBench (not shown), the method’s resolution fails: with mostly binary answers the per-item split ($n = 57$, 16 skills) gives the transplant 0.75 [0.60, 0.88], its own d on the shared prefix 0.80, half a dose 0.44, another skill 0.38, permuted 0.25 and noise 0.15, and the depth curve has no cliff (0.93 at layer 0, 0.49 at 16, 0.33 at 20).

C.3 LIMITATIONS, IN FULL

Sample size and scope. The battery was developed on $n = 40$ (ten calculators $\times$ four rescued instances), which gives intervals of roughly ± 0.15 ; §4 reruns it on forty calculators and §5.3 on further models and a second task, and the conclusions are the ones that survived that. We report profiles and not individual layers: the one non-monotonicity in the depth curve (0.250 at layer 18 against 0.475 at 20) has overlapping intervals and we do not interpret it. Among the three models, Qwen3-1.7B is the weakest rung: the document rescues only 8–11% of its items, so its truncation curve is not resolved and is reported for completeness rather than relied on. The width argument rests on Qwen3-0.6B against Qwen3-8B, a factor of four within one family; a second family of the same width needs more directions (Appendix C.2). Four independent shards recomputed the three baselines and agreed to the item (0.950 / 0.025 / 0.175), and the no-op arm sits on the receiver’s baseline at all twelve layers to within the 5% numerical floor of §2, so the instrument is checked even where the estimate is coarse. The span channel, the depth sweep and the truncation curve have three models and two tasks; the knockouts two models and two tasks; the final-position results two models (Qwen3-1.7B, fp32); the document-order comparison and the within-document localisation one.

Which items the causal claims are about. Two restrictions stack. Causal claims are made only on the rescued cell, the only cell in which the document demonstrably works; on the persistent-failure cell the same arms track the document rather than correctness—the transplant fails exactly where the document fails—which is the right behaviour but is not a substitute for the other cells. And within that cell the span arms are read on the calculators whose *receiver* never solves an item, because on the others the model does not need the document, and before the fix any disruption of the wrong one, noise included, let it fall back on what it already knows ($\rho = 0.14$ for noise there against 0.00 here). After the fix noise no longer does that (0.00 on those calculators), but a donor from another skill still reads 0.47 there against 0.19 on the rest. On forty calculators the restriction keeps fifteen of them and 60 items, so the intervals are no longer the binding problem; what the restriction buys is that the comparisons inside it are between 0.750 and at most 0.083 rather than between numbers a confound could produce.

What the rank result does and does not establish. §5.2 truncates the injected content matrix to rank k at a layer where the untruncated version recovers most of the effect, and the effect is absent at $k \leq 32$ while the bottom- k control stays at the floor. That licenses the forward direction: the effect requires directions of the same order as the span’s participation ratio, and within the Qwen family neither number scales with width. It does not license a claim that the number belongs to the document alone (Mistral-7B needs more), nor a converse about depth. Supplying a deep layer with a higher-rank message taken from a shallow one does not restore the handover ($\rho = 0.00$ against 0.06 for the same-layer donor, 0.00 with the norm matched), so we do not claim that what a deep layer lacks is rank.

Two negative results, stated as limitations. We could not find any representational quantity at the final position that distinguishes the items a skill rescues from the ones it does not: the norms and angles agree to three digits, and the one similarity measure that separates them under the gold skill separates them equally under a wrong one. And the document acts through a chain of thought, so every causal measurement decodes several hundred tokens; scoring under a forced ANSWER: cue instead removes the behavioural effect entirely (0.175 for all three arms) while leaving a +0.60 nat log-probability difference—so the span channel has no usable log-probability reading, and is reported on accuracy alone.

An instrument fault, and which results predate its fix. The state capture called the model without an explicit attention mask while decoding passed one; for a single unpadded sequence these are

mathematically identical but take different attention kernels, and they disagree by 1.2% of the state’s magnitude at layer 8 and 25% at layer 34. The visible consequence was that the null arm—the receiver’s own states written back over themselves—changed the answer on 6.1% of Qwen3-8B items. Passing the mask in both makes the null arm exact (80/80 across two reruns) and leaves the transplant at $\rho = 1.00$ as before. *Rerun after the fix and reported from the reruns:* Table 1 including the same-family donor, the forty-calculator sufficiency curve, the layer windows, the forty-calculator rank curve, the ten-calculator dose sweep and battery (Figure 1), the document-order comparison at layers 8 and 16, the within-document quarters on ten and forty calculators, the Qwen3-0.6B truncation curve, and every TheoremQA and Mistral-7B run; the attention knockout does not use the capture path. *Still from before the fix:* the synthetic tier and the final-position sweeps, the ten-calculator development rank curve, the rank curves at layers 12 and 16, the cross-layer donor, the Qwen3-0.6B depth sweep and knockout on MedCalc, Qwen3-1.7B, the halves split, and Figures 3 and 4; these are read against a null arm at 0.034 rather than 0.000 on the restricted split. Where a rerun changed a conclusion—the worked example’s share of the document, the instance-specific part at layer 16, the dose curve’s shape—the text says so.

Numerics. Log-probabilities on the synthetic control are fp32, because bf16 costs 0.5–2.5 nats on a token deep in the tail. The 8B numbers are bf16, which appears as self-transplant deviations of 0.1–1.1 nats while the accuracy channel remains an exact no-op; accuracy is read first throughout.

C.4 WHERE IN THE NETWORK THE CONTENT IS READABLE, IN FULL

Transplanting one group of positions from the with-document forward into the without-document forward asks where the content can be *read from*; masking attention to the document asks where it is still being *read*. In the multiple-choice format the first question gives three windows that hand off cleanly (Figure 9, Appendix C): the document’s own 688 tokens work at layers 0–10 and fall off a cliff at 11, the question’s 42–47 tokens work at 11–16, and the final prompt position works at 21–27. The handoff is not an artefact of a shared analysis—the two experiments were run independently—and at layer 11 the document-span channel drops from 0.308 to 0.128 while the question-span channel rises from 0.231 to 0.308. The windows survive a change of model and not a change of format (Table 11): in free-form, on the same model and items, the question span recovers 0.000–0.025 at every one of 12 layers.

The second question gives a different shape. No single pathway is necessary: masking only the final position’s attention to the document costs nothing (0.436 against 0.436) and masking only the question span’s leaves 0.282, so the read is redundantly distributed, as Li et al. (2025) report for task representations generally. What is necessary is the document’s own tokens. On third-party skills, making them unattendable for the whole forward takes accuracy from 0.550 [0.400, 0.700] to 0.075 [0.000, 0.175], against 0.025 [0.000, 0.075] with no document in the prompt at all—ninety per cent of the benefit, with the remainder inside the no-document interval. Masking an equal-length span of the patient note instead costs less, 0.150 [0.050, 0.275]. A genuinely inert length-matched control cannot be built here, since the only other long span is the data the task needs, so this bounds “removing 900 tokens” loosely but in the right direction.

Single-layer knockouts measure nothing on a 688-token span, because a layer denied the read recovers it one layer later; cumulative knockouts are required and are what we report. Cumulatively, the document stops being needed at layer 22 of 28 when the answer is one token, and is still being read at 83% depth when the answer is several hundred—the necessity-side view of §3.1.

C.5 THE DECOMPOSITION AND THE ANSWER-LENGTH LADDER, IN FULL

We add each component back to the no-document state at the last prompt position and let the forward pass continue, $h' = h_{\text{no}} + \alpha v$ for $v \in \{d_i, g_i, t_i, \bar{d}, d_j\}$, against an upper reference that replaces the state outright with h_{skill} . The protocol needs a task with an exact zero floor and a matched negative control, so it is established on the synthetic tier and then carried to the clinical one; Appendix C.18 gives the full sweep, and this is the only place in the paper where the synthetic material carries an argument of its own.

Three results, and the third is the one the rest of the paper uses. **The presence component re-pairs nothing but is not inert:** injected alone it takes the rescued cell to 0.062, and putting the

wrong document in the prompt also gives 0.062—it faithfully reproduces the behaviour of having a document, which is worse than having none. **The content component repairs most of the cell**, 0.812 against an upper reference of 1.000, having been dismissed by every geometric measurement as a small residual. **The patch reproduces the document’s whole behavioural signature**: across the four cells, replacing the position with h_{skill} gives 1.000/0.000/1.000/0.000 and adding d_i gives 0.812/0.112/1.000/0.125—the same signature, attenuated. A patch that only ever helped would be consistent with a generic “try harder” effect; one that also breaks exactly the items the document breaks is carrying the document’s decision.

Which control you subtract chooses what you measure. d is defined against a control document, and the choice is a measurement decision rather than a detail. Against an *unrelated* document of the same length it is topic, procedure and numbers together and repairs 0.812 of the rescued cell; against the *same* document with every number replaced it is the numbers alone and repairs 0.229–0.312 (layers 21–27); against the same document with its sentences shuffled, 0.188–0.292. The three differ by a factor of three to four, so a paper that reports “the content component” without naming its control has not specified the quantity. Appendix A gives the full comparison.

C.6 WHAT A SINGLE POSITION CAN CARRY IS THE ANSWER, NOT THE CONTENT

Single-position patching is the standard move in this literature, and Table 23 is a multiple-choice result, so it is worth asking what that one position is carrying. The same items exist in a free-form numeric form, where the answer is a number the model must compute rather than an option it must select; there the skill is worth $0 \rightarrow 0.467$ and the rescued cell has 56 items.

Two things change and one does not. The ceiling collapses: through layer 22 every condition reads zero, including replacing the position outright with the true skill state, and the best any single-position intervention achieves is 0.304, at layer 32. What does not change is the ordering—the content component recovers 0.268 of that 0.304 and the presence component 0.018—so the conclusion of §3 survives the format while the magnitude does not (full sweep in Appendix D).

It is the format, not the model. The identical sweep on the *same model and the same items* in multiple-choice form (Qwen3-8B, $n = 120$, $n_R = 44$) recovers 0.977 at layer 26 and 1.000 at layers 30 and 35. Same model, same items, same intervention, same layers: 1.000 when the answer is one option letter, 0.304 when it is a number the model has to produce.

The reason is mechanical and can be counted. Of the 120 items, *none* has a gold answer that is a single token under this tokenizer: 36 are two tokens, 55 three, 29 four. A patch at one prompt position can fix at most the first generated token; the remaining one to three are produced in a context that still does not contain the conversion table. Multiple choice compresses the answer to exactly one token—which is exactly the unit a single-position patch transports. Patching the last k positions instead of one confirms that it is a capacity and not a threshold: $k = 1, 4, 16$ recover 0.286, 0.286 and 0.554 at layer 30, so sixteen positions carry about half of what the document carries and four—the chat suffix—carry nothing extra.

On third-party skills the same probe reads nothing. There, replacing the final prompt position with the with-skill state repairs at most 3 of 18 rescued items (layer 10) and 0–1 from layer 14 on and transplanting the question span—50 to 1697 positions, into a length-matched receiver—repairs nothing beyond the receiver’s own baseline at any of seven layers (Table 12), with the self-transplant arm on that baseline to the item, so the instrument is live and the null is real. §4 shows what does work on the same material.

Answer length is the cause, and that is a controlled comparison. The step from the synthetic tier to the clinical one changes the skill, the task, the prompt length and the fraction of it a patch covers, so it cannot on its own identify what makes the injection fail. We therefore made answer length the only thing that moves: the *same* 80 synthetic items, the same model, the same document, prompted to reason step by step and give the answer on a final ANSWER: line. In that format the skill is worth $0.000 \rightarrow 1.000$ —every item is rescued, $F = K = B = 0$ —so there is more effect to recover than in either other format, and the rescued cell is the whole item set.

The single-position patch recovers 0.000 of it (§3.1), as do d and g . The last-layer instrument check passes on 10 of 10 items in the same run.

The range over which “a skill’s effect is carried by a vector” is true is the range over which the answer is short—and the clinical result is then not a property of that material but an instance of this.

It is a capacity limit, and it can be measured. Patching one position is a choice, not a measurement, so we repeated the sweep patching the last k prompt positions for $k \in \{1, 4, 16\}$. Replacing them with the skill state recovers 0.286, 0.286 and 0.554 of the rescued cell at layer 30. Four positions buy nothing—the last four tokens are the chat suffix—and sixteen, which reach back into the question, roughly double the recovery. Against the document itself at 1.000, even sixteen positions carry about half. The content does not compress into a small set of positions; how much of it a patch transports is a function of how many positions it is given.

We therefore read the low ceiling as a scope condition on single-position patching rather than as a negative result about d . In a multiple-choice item the answer *is* one token at one position, so a patch that transports the model’s decision transports the whole task and looks like it transported the document. What the final prompt position holds by layer 21 is the model’s *answer*, not the document’s content—consistent with the knockout result that reading the document is finished by that layer (§5).

C.7 SETUP, IN FULL

Two task families. *Real skills, the main material:* MedCalc-Bench instances with the 55 gold calculator skills released with SRA-Bench—55 calculators $\times$ 20 instances = 1,100 items, graded deterministically against a numeric tolerance with no model in the loop. Both the skills and the tasks are other people’s, which is what makes the effect size credible. Qwen3-8B, single turn, no tools, greedy; intervals bootstrapped over *calculators*, since the 20 items sharing one are not independent. *A synthetic positive control:* a fictional unit system with a one-page document giving its conversion table, so the no-document floor is guaranteed, contamination is impossible, and we know exactly what content means. Items are generated from the table ($n = 358$), and every item whose answer appears in a worked example is dropped.

Two dependent variables. Accuracy, from a greedily decoded free-form answer graded by the task’s own scorer, and the teacher-forced log-probability of the gold answer. They fail differently and disagree often enough that reporting one alone is unsafe (§7); accuracy is read first throughout. No item was rewritten into multiple choice to obtain an accuracy channel—§C.17 shows why that rewrite is not harmless.

How to read an arm’s number. Interventions are reported both as raw accuracy and as the fraction of the range an intervention could possibly recover,

$$\rho(\text{arm}) = \frac{\text{acc}(\text{arm}) - \text{acc}(\text{receiver})}{\text{acc}(\text{donor}) - \text{acc}(\text{receiver})}, \quad (5)$$

where the receiver is the prompt being patched into, unpatched, and the donor is the document actually in the prompt. ρ is what transfers across settings and raw accuracy is not: the four configurations in Figure 5 have donors at 0.550, 0.467, 1.000 and 0.950 and receivers from 0.000 to 0.350, and an earlier draft of this paper reported a channel as *stronger* under a longer answer by comparing 0.408 against 1.000 when as fractions they are 0.87 and 1.00. Every figure axis labelled “fraction” is Eq. 2.

The instrument’s own noise floor, measured. Every decoding run carries a *null* arm: the receiver’s own span states are captured and written back over themselves. With a length-matched receiver this is a bit-for-bit identity, so the null arm must equal the unpatched receiver at every layer. Before the fix of Appendix E it did not, quite. On Qwen3-8B, across 1,766 paired observations from every pre-fix run that carries both arms, the null arm agrees with the receiver on 93.9% of items: 108 flip, 46 from wrong to right and 62 from right to wrong, symmetric within a sign test ($p = 0.15$). The pipeline itself is deterministic—the same configuration run on two different machines agreed on all 360 item-by-arm cells—so this is not run-to-run variation but a reproducible property of the

instrument: the capture and decoding passes took different attention kernels, which perturbs the written-back state by 1.2% at layer 8, and the dependent variable is the end of a chain of thought several hundred tokens long, which is long enough for that to change an argmax somewhere and propagate. It scales with the model: the same arm flips 0.6% of items on Qwen3-0.6B (1,696 pairs) and 0.8% on Qwen3-1.7B (1,338), against 6.1% on the 8B. After the fix it is exact (80/80).

Three consequences are used throughout. First, **an effect smaller than about six points is not resolvable on the 8B** in pre-fix runs, whatever a binomial interval says. Second, the floor is not uniform: on the persistent-failure cell it is 16%, between two and three times the average, which is what one expects of items on a decision boundary, so numbers read on that cell are read against a coarser ruler. Third, on the restricted split the pre-fix null arm read 0.034 rather than zero—the split conditions on the receiver failing, so a flip there can only go upward—so pre-fix span numbers are read against 0.034; after the fix it reads exactly 0.000, the floor Table 1’s 0.750 is read against. We have rarely seen this control reported in the patching literature; at a single answer token it is small enough to be invisible, and it becomes large only once the dependent variable is a long generation.

Four cells. Every item is labelled by what the document did to it: **R** (rescued: wrong without, right with), **F** (persistent: wrong both ways), **K** (kept: right both ways), **B** (broken: right without, wrong with). **R is the only cell in which the document demonstrably works**, every table here is read by cell, and §7 shows the aggregate over all four can be sign-inverted relative to R.

C.8 THREE MEASUREMENT TRAPS, IN FULL

C.9 THE AGGREGATE GOLD-ANSWER LOG-PROBABILITY RANKS THE CONTROLS FIRST

Averaged over all items, the log-probability of the gold answer under each patched condition at layer 27, the last, orders as

mean vector -2.85 > filler state -5.20 > **true skill state** -6.60 > no document -7.30 > wrong skill -10.29 ,

i.e. a content-free average vector scores 3.75 nats *better* than the state the model actually builds while reading the document, with accuracy 0.256 against that state’s 0.436.

The cells of §2 account for the gap arithmetically (Table 13): F and B—where the document does not work—contribute 133% of it at layer 21, where the gap is 2.75 nats, while on R the true state wins by 2.5 nats and 0.69 in accuracy. The mechanism is visible in the option distribution: the mean vector raises the entropy over the four answer tokens from 0.187 to 0.972 nats (maximum $\ln 4 = 1.386$). A flattened distribution cannot assign any token -19 nats, so it looks excellent exactly where the true state is confidently wrong, and flattening never promotes a token to the argmax, so accuracy does not move. The one point the two channels agree on is that the *wrong skill* is worst on both—positive evidence for content specificity, since injecting the wrong content actively misleads.

The practical rule is that an aggregate log-probability over a mixed item set measures confidence calibration as much as it measures content, and that a decomposition by outcome cell is the cheapest way to tell which.

C.10 A BAND WHERE THE FORWARD PASS IS DAMAGED

Between layers 17 and 20 every patched condition improves together and the conditions become indistinguishable. Reading by cell isolates the cause: on the K cell—items the model already answers correctly—the true-state patch drops the gold log-probability from -1.6 at layer 16 to -7.1 at layer 18, while the filler-state patch, built over an equally long prompt, stays at -1.5 . Any transfer or recovery number falling in 17–20 is reported with that band marked.

C.11 WHAT THE WRONG-DOCUMENT CONTROL IS ACTUALLY WORTH

Holding the prompt’s length, structure and the document slot’s position fixed and changing only what sits in that slot gives an ordered comparison the usual binary—document or no document—cannot

make. On the rescued items, with everything else identical ($n = 40$, intervals in brackets): no document at all 0.025 [0.000, 0.073]; non-clinical prose in the slot 0.250 [0.116, 0.384]; this skill’s own paired neutral document 0.175 [0.057, 0.293]; one fixed wrong clinical skill 0.150 [0.039, 0.261]; the gold document 0.950 [0.882, 1.000].

Two things follow, one solid and one not. The solid one is that the wrong-document control is not the same object as the no-document control: any document in the slot moves these items off the floor, so a paper that reports “with the document” against “without” is measuring presence and content together, and the presence part is not negligible on a cell selected for failing without. That is the reason every arm in §4 is measured against a receiver that holds a document rather than against an empty slot.

The one that is not solid is the ordering. A clinical distractor costing more than irrelevant prose (0.150 against 0.250) is the direction we expected—a near miss should be worse than an obvious miss, and the three conditions fall in that order—but every pair of intervals overlaps at $n = 40$, so we report a direction and not a result. We flag it because the design that would settle it is cheap and this material affords it: these 55 skills contain graded near misses by construction (four different formulae for the corrected QT interval, two for the delta gap), so “how similar must a wrong document be before it hurts” can be measured on an ordered scale rather than a binary. We have not run that experiment.

C.12 DEPTH, SUFFICIENCY AND NECESSITY, IN FULL

Table 6: Participation ratio of the document’s span on Qwen3-8B, measured four ways, medians over 36 items (the plain column is bimodal, so its mean describes no item). The first column is the quantity usually reported and the only one that collapses. The last two columns say why: at layer 16 one token in 34 of 40 documents acquires a state 150 times (median) the span’s norm, carried by two fixed hidden dimensions, and the top eight dimensions then hold 98% of the variance. Removing four positions out of four hundred, or normalising each position to unit length, removes the collapse entirely.

layer	participation ratio, computed. . .				massive activations	
	as usual	drop 4 loud	unit-norm.	drop 8 dims	max /med $\ h\ $	top-8 var.
8	66.0	67.2	65.3	75.3	1.5	4.9%
12	48.9	53.8	53.0	57.7	1.6	5.6%
14	55.4	58.1	57.9	64.8	1.6	5.4%
16	1.0	65.9	67.6	26.9	150.8	97.7%
20	1.1	73.6	73.8	57.5	94.0	94.9%
28	2.4	53.9	59.6	75.4	26.8	65.4%
34	27.2	27.6	55.6	82.0	1.3	16.1%

On the clinical material the two questions—*can* the document’s states be handed over at layer L , and *is* the model still reading them at layer L —give different answers, and the disagreement is the informative part.

Sufficiency, on forty calculators after the instrument fix ($n = 60$, twenty calculators after the restriction): a single layer’s worth of the span reproduces $\rho = 1.04, 0.94, 0.85, 0.77$ at layers 0, 4, 8, 12 and then 0.28, 0.19, 0.19, 0.15 at 14, 15, 16, 20. The first, pre-fix sweep on twenty items (Table 8) has the same shape: 1.00 at 0 and 12, three quarters at 13–14, 0.14 at 15 and 16, nothing by 24.

Necessity, from cumulative attention knockout with the full chain of thought decoded ($n = 40$ rescued items, twenty calculators): masking every position’s attention to the document’s tokens from layer L to the end leaves 0.08–0.15 of the items correct for every $L \leq 20$ except $L = 4$ (0.33), then 0.38 at 24, 0.75 at 28 and 0.95 at 32. An earlier knockout run capped generation at 400 tokens, truncating the chain of thought so that even the unmasked prompt scored 0.50; it is superseded and not used. The model is still reading the document long after a single-layer handover has stopped working.

It is not that one layer is too few: the channel closes. The natural reading of the cliff is that a single layer stops being enough, since writing the span at *every* layer at once still recovers 0.950.

It is wrong, and one experiment separates the two. We write the span at every layer in a window $[\ell_{lo}, \ell_{hi}]$ and nowhere else. Before the instrument fix, on $n = 27$ items over seven calculators:

layers written	{16}	16–17	16–19	16–23	16–35	14–19	12–15
how many	1	2	4	8	20	6	4
ρ	0.07	0.07	0.11	0.04	0.04	0.44	0.96

Rerun after the fix on twenty calculators ($n = 48$), the windows read 1.00 for 8–11, 0.96 for 12–15, 0.38 for 14–19 and 0.11 for 16–35, against a null arm at exactly 0.00. Four layers are enough if they are layers 12 to 15. Twenty layers are worth almost nothing if they start at 16. Widening the window from one layer to twenty buys nothing at all. Whatever the document’s span is for, the model is done with it by layer 16, and after that the positions are no longer a route into the computation—not a narrow route, not a route that needs more layers, no route. The window 14–19 sits at 0.38–0.44 because it straddles the boundary.

This is worth stating as a negative constraint on any mechanism, including the one we tried to give it. Whatever ends at layer 15 is not a capacity that more depth can compensate for.

The span’s representation does *not* become a summary, and the measurement that says it does is an artefact. The obvious explanation for the cliff is that the span has by then been compressed, and the obvious measurement agrees: the centred participation ratio of the span’s states, which counts the directions its positions differ in, reads 64 at layer 12 and 17 at layer 16—the same interval. We report it here because we believed it, built a mechanism on it, and it is wrong.

Three things give it away. It is a step and not a slope: the span states’ participation ratio lies between 47 and 75 for every item at layer 14 and below 5 for thirty-four of forty at layer 16 (Table 6), and the content matrix’s mean of 17 at layer 16 describes no item—thirty-two of forty read below 5 and the other eight at 81. A participation ratio of 1.0 over four hundred positions asserts that the centred span is rank one, which is not a statement one can make about four hundred tokens of text that the model still has to continue. And the collapse is absent in Qwen3-0.6B and Qwen3-1.7B at every depth (45–99, no item below 5) although their transplant cliff is in the same place in relative depth (§5.3).

The cause is measurable and known. At layer 16—and at no earlier layer—one token in 34 of 40 documents acquires a state whose norm is 99 to 180 times the span’s median, carried by hidden dimensions 2276 and 676, the same two in all 34. That is the massive-activation phenomenon of Sun et al. (2024), which appears at a single layer per model family (Shi et al., 2026); in this model the loud token is a late delimiter inside the document rather than the sequence start. A centred spectrum is then owned by that one position: the top eight dimensions hold 97.7% of the variance at layer 16 against 5.4% at layer 14. Removing the four loudest positions out of four hundred restores the participation ratio to 66; unit-normalising each position restores it to 68; removing the top eight dimensions gives 27. Measured any of those three ways, the span’s dimensionality is *flat* from layer 0 to layer 28 and only falls in the last few layers (Table 6).

So the honest statement is the narrower one. The single-layer handover dies between layers 14 and 16; the document’s representation does not lose dimensionality there; we do not know what it does lose. What §5.2 establishes is a different and still causal fact—how many directions the injected object needs—and in the small pre-fix repeats at layers 12 and 16 that number is not lower either (Appendix C.1).

A warning about participation ratio on residual streams. Effective rank, participation ratio and intrinsic dimension are routinely computed on residual-stream activations, including in work that reports collapses with depth (Fernando & Guitchounts, 2026; Sarkar & Deka, 2026). Massive activations begin at a fixed layer, are carried by a handful of dimensions at a handful of positions, and are large enough to own a spectrum by themselves. Any such measurement that does not exclude them will report a collapse at exactly the emergence layer whether or not the representation has changed. Our own draft did. We recommend reporting the per-position norm ratio alongside any spectral statistic; here its median goes from 1.6 to 150 across the layer in question.

C.13 WHAT THE CONTENT COMPONENT BELONGS TO, IN FULL

If d were a “skill vector” it would be reusable: extracted from one task it would work on another. Its *direction* is reusable. Its *effect* is not, and the gap between those two statements is the finding.

The vector that carries contains no task information, and we check rather than argue it. Everything in this section so far is about the vector at the final position, the one that does not act. The vector that does act—the content component over the document’s own span—has a property that makes the question sharp. In the prompt format the behavioural runs use, the document precedes the question, so causal masking makes the document’s hidden states a function of the system text and the document alone. Within one skill the document is byte-identical across its instances, so $d(\text{span})$ should be *the same tensor* for every instance of that skill.

It is. Across the ten calculators in the causal set, the document span occupies token positions $43 + m$ in every prompt, the token ids there are pairwise identical across instances (zero differing positions in 40 notes), and truncating two prompts to the end of the span and re-running the forward gives a relative difference of 0.0. In the full prompts the states differ by a relative 3.5×10^{-3} at layer 0 and 8.6×10^{-3} at layer 8, with zero positions below cosine 0.999—accumulated bf16 rounding from the differing sequence lengths, not content. The decode arms show the same thing behaviourally: the mean of the *other* instances’ vectors and the instance’s own agree to three decimals at every running mean, and the component of d_i orthogonal to that mean lands exactly on the receiver’s baseline.

So the vector that reproduces the document’s entire effect is a function of the document and of nothing else. Whatever a “skill vector” should be, this one is not a joint encoding of skill and task; it is the skill, and it is sufficient. The complementary experiment—moving the document behind the question so that its states *can* see the task—is §C.15.

The direction differs by skill; the effect does not transfer. On the clinical material each skill has 20 instances, so “same skill, different task” is directly measurable. Content vectors from different patient notes under the same calculator have cosine 0.789 at layer 20 and 0.876 at layer 30; vectors from different calculators, 0.320 and 0.653 ($n = 40$, Qwen3-8B). On the synthetic tier the analogous split is by conversion family, 0.859 within against 0.699 across at the last layer. The direction of d carries the identity of the document, increasingly so with depth. Its effect at the final position does not: injecting the mean direction $\bar{d}$, or another item’s d_j , repairs 0.000 at every one of 16 layers in the free-form format against 0.268 for the item’s own, and in multiple choice reaches 0.062, 0.292 and 0.167 for the mean, a same-family and a different-family donor against 0.792 for d_i (Table 23). The one band where a donor looks transferable is the band §7 identifies independently as one where the forward pass is damaged and every condition rises together.

Nothing about it distinguishes rescued items from persistent failures. The norms and angles of d and g agree to three digits across the two cells (layer 30, 8B, real skills: $\|d\|$ 189.4 vs 185.4, $\cos(d, g)$ -0.469 vs -0.468). A representational-similarity measure over the question span does separate them under the gold skill at layers 10–16 (-0.0037 to -0.0070 , non-overlapping intervals, $n = 150 + 150$)—and separates them just as much under a length-matched *wrong* skill (-0.0035 to -0.0056). Zero layers pass a control the design required in advance, so what looked like a signature of “the skill worked here” is a signature of “a document is present”: §B’s trap one register down, and the reason we report a causal channel rather than a similarity score.

The content component’s size depends on where it is written. At the final position the content component is a 6% perturbation of the state it displaces at layer 6 ($\|d\| = 2.3$ against $\|h_{\text{no}}\| = 37.3$), rising to 51% by layer 35—and it is causally inert at every one of them. Over the document’s own span the same difference of the same two conditions is 95% to 115% of the state at every depth (Table 9)—a replacement rather than a nudge—and it is causally complete (§4). What distinguishes the two places is not how much the document is written into them; it is whether what is written is the whole of what the downstream computation reads.

The shared part carries the norm, not the effect. Between 73% and 99.7% of each item’s d_i lies along the shared mean direction, depending on layer (Table 22); the remainder occupies an effective 3–18 dimensions out of 2048–4096. The reusable part is almost all of the norm and, in free-form

answering, none of the effect; in multiple choice it is worth 0.06 of the 0.79 the item’s own vector achieves. A single vector per skill is therefore a fair description of what the injection *looks like* and a wrong description of what it *does*, which is the same inversion as §B one level down.

C.14 THE SPAN BATTERY, IN FULL

§3.1 is a negative result about a probe, not about a model. The probe fails because one position cannot hold a decision that occupies hundreds of tokens. The obvious repair is to stop using one position, and the natural place to use instead is the one group of positions whose size grows with the document: the document’s own tokens.

The receiver. The donor is the with-document forward; the receiver is the *same prompt* with the document’s tokens overwritten, one for one, by the tokens of *one fixed clinical skill document*—the same document for every instance and every skill, and the gold document of none of them.

Both halves of that choice were forced by measurement. It has to be fixed, because a receiver built from each skill’s own paired control makes the receiver’s states depend on the instance, and then every arm of the form $h_{\text{recv}} + (\cdot)$ inherits an instance-dependent floor; measured, the with-document states vary across a skill’s instances by a relative 3.5×10^{-3} while such a receiver’s vary by 0.36. We use a *clinical* document rather than neutral prose because it is the same condition the behavioural runs call “wrong skill”, so the receiver’s baseline means what the paper’s presence effect means; §C.11 measures the two side by side and finds them close (0.150 against 0.250, $n = 40$, overlapping intervals).

Length-matching is not a refinement here either, it is the whole experiment. An earlier version of this run used the no-document prompt, which is some 900 tokens shorter, and aligned the spans by their ends; the real arm and the wrong-document arm then both read 0.200—the signature of two arms broken in the same way by a rotary phase shift, which is easy to read as an absent effect. With a matched receiver every position keeps its index, the two sequences are identical outside the span, and the question span is verified token-identical between donor and receiver before anything is patched.

Because the receiver already holds a document, the substitution obeys $h_{\text{recv}}(\text{span}) + d(\text{span}) = h_{\text{skill}}(\text{span})$: it is $h_{\text{recv}} + d$ at $\alpha = 1$. That is what makes the rest of this section possible.

It carries, completely. Figure 1 puts the two channels side by side on the same model and the same decomposition. At the last prompt position, five arms—including replacing the entire state with the with-document one—sit between 0.050 and 0.150 against a no-document baseline of 0.050. Over the document’s own span, after the instrument fix, the same content vector gives 1.00 against a receiver at 0.00 ($n = 16$, four calculators).

Before the fix the same panel read 0.950 on twenty items, $\rho = 1.06$: the transplant and the document disagreed on one item and the transplant got it right. The honest reading of every ρ near or above 1 here is “indistinguishable from having the document in the prompt”.

How deep the handover works (before the fix). Transplanting the *document’s own* states over the same construction recovers the document’s entire effect through layer 12 and then falls off a cliff (Table 8). Layer 0 is close to tautological—a layer’s output over the document’s tokens is not far from putting those tokens back—so the load-bearing entries are layers 8 and 12, a third of the way into the network, where the receiver has already processed eight or twelve blocks’ worth of filler in those positions and the transplant still recovers everything. Past layer 14 a single layer’s worth stops being enough: $\rho = 0.14$ at 15 and 16, and 0.00 at 24 and 30. Writing the span at *every* layer at once returns it to $\rho = 1.00$, so what fails past layer 14 is the handover at one depth, not the mechanism.

The profile replicates; its one anomaly does not. We ran the sweep a second time on the same items against the other receiver—the single fixed wrong skill rather than each skill’s paired control—and the two agree wherever the first is well behaved: +1.00 against +1.00 at layer 0, +1.06 against +1.07 at 4, +1.03 against +1.14 at 8, +0.81 against +1.00 at 12 (Table 16). The single bump in the first run, $\rho = 0.07$ at layer 18 against 0.29 at 20, does not survive: the replication gives +0.16 at

layer 20, below its own layer 16. We read the profile and not the individual layers, and we read that bump as noise.

The surviving window is a window in *relative* depth: layers 0–12 of 36 here, 0–10 of 28 on the synthetic tier, both about a third of the way in.

It behaves like a mechanism, not like a perturbation. Four controls, all at layer 8, all in Figure 1, and all read on the calculators a wrong document never rescues.

Dose: after the fix, $\alpha = 0.4, 0.5, 0.55, 0.6, 0.7, 1, 2$ give $\rho = 0.06, 0.00, 0.06, 0.62, 0.94, 1.00, 0.81$ ($n = 16$). The transition is a step and not a ramp—half the content is no better than none of it, since the state is then a midpoint between two coherent documents and corresponds to neither—and over-shooting costs a little. An injection that merely had to be large would show neither. Before the fix the same sweep read 0.17, 0.03, 0.17, 0.88, 0.97, 1.00 ($n = 30$), and a coarser one 0.06, 1.06, 1.06, 0.89 at $\alpha = 0.5, 0.75, 1, 2$ with the per-position rescaled doses coinciding (0.06 and 0.89 both ways).

Transfer: after the fix, another skill’s vector over the prefix the two documents share gives 0.19 against 0.94 for this skill’s own vector on the same positions ($n = 16$), and on forty calculators a donor from another family 0.06 and one from the same family 0.06 against 0.73. Before the fix, the mean content vector of a *different* skill, written over the prefix the two documents share, gave $\rho = 0.06$ against 1.00 for this document’s own vector written over *the same positions*. Because every skill’s vector is taken against the same fixed control, this arm is exactly that skill’s own span states rather than a hybrid, so it is the cleanest form the transfer test can take; and the paired control is what makes it a statement about content rather than about how many positions were patched.

Order: the same vectors, permuted across the span’s own positions, give $\rho = 0.00$ ($n = 16$) and 0.08 on forty calculators after the fix (0.12 before). The span is not a bag—which position holds which part of the document is most of the effect.

Norm: Gaussian noise rescaled to each position’s $\|d\|$ gives $\rho = 0.00$, exactly the receiver (0.10 on forty calculators). The effect is not a consequence of writing something large into those positions.

The split these four are read on, and why it is not cherry-picking. On four of the ten calculators the receiver—a wrong clinical document—already solves the items. Before the fix every span arm worked there: a related-skill donor scored $\rho = 0.71$, an unrelated one 0.43 and norm-matched *noise* 0.14, against 0.00, 0.06 and 0.00 on the five where the receiver scores zero, and pooling the two groups turned a null transfer arm into an apparent $\rho = 0.38$. After the fix six calculators fall on that side; noise no longer works there (0.00, $n = 24$) but another skill’s vector still reads 0.47 against 0.19 on the four restricted calculators: there the wrong document in the slot was leading the model astray, and a coherent replacement lets it fall back on a formula it already knows.

The split conditions on the *receiver*, which is a control, and on no intervention arm, so it cannot manufacture the contrast it is used to report; and it is taken per calculator rather than per instance because “this model can do this calculator unaided” is a property of the calculator. Table 17 gives every arm on all forty items for comparison.

It replicates on four times as many skills. The ten calculators above are the ten with the most rescued instances, and twenty items from five calculators is the single most obvious reason to stop reading. We therefore ran the battery on *forty* calculators, recomputing the restriction inside the larger set rather than carrying it over. Of the forty, fifteen are calculators the wrong document never rescues, which leaves 60 items—three times the original—and every arm holds (Table 1): the document’s own vector recovers $\rho = 0.87$, the same vector over only the prefix another skill’s document shares 0.73, half a dose 0.06, a donor from another family 0.06, permutation of the span’s own positions 0.08, and norm-matched noise 0.10, against a null arm at exactly 0.00 and a receiver at exactly 0.000, and a donor from the same family 0.06 (rerun after the instrument fix). The separation is between 0.87 and at most 0.10, on three times the material, with intervals bootstrapped over calculators.

And it fails where the document fails (before the fix). On the persistent-failure cell—items the gold document does *not* rescue—the same transplant at the same layer gives 0.100, equal to the receiver’s 0.100, against the document’s own 0.025 ($n = 40$, ten calculators; the null arm read

0.150 in this run). The channel transports what the document does, including doing nothing. An arm that only ever helped would be consistent with a generic “try harder” effect; this one is not.

Why it can carry when a single position cannot. The span is 415 to 1,630 positions wide on the forty calculators, the same order as the answer it has to support. This is the positive half of §3.1’s capacity statement, and the two together give the rule the paper is built on: *a patch transports what fits in the positions it is given*. A one-token answer fits at one position; a chain of thought fits nowhere except on a group of positions as large as the document itself.

Table 7: Behaviour on the real skills, Qwen3-8B, all three arms on the same 1,100 items, CIs bootstrapped over the 55 calculators. The presence effect (wrong skill minus no skill) is a clean zero, so the +39.7pp is content.

	<i>n</i>	accuracy	95% CI
gold skill	1100	74.2%	[65.8, 81.8]
no skill	1100	34.5%	[26.5, 42.8]
wrong skill (length-matched)	1100	31.5%	[23.7, 39.5]
gold – no skill	1100	+39.7pp	[+31.7, +47.7]
gold – wrong skill	1100	+42.6pp	[+34.7, +50.5]
presence: wrong – no skill	1100	–2.9pp	[–8.8, +3.0]

Table 8: The document-span transplant by layer. Qwen3-8B, 55 third-party clinical skills, the 40 rescued items, the full chain of thought decoded and graded by the task’s own scorer. The receiver is the with-skill prompt with the document’s tokens replaced by a filler document, before the instrument fix, so it is the same length and every transplanted position keeps its rotary phase. Restricted, as everywhere in this section, to the calculators the receiver never solves. This sweep used each skill’s paired control as the filler and the battery of §4 the single fixed one; over all forty items the three baselines agree to within 0.01 across the two (0.175/0.025/0.950 against 0.167/0.028/0.944), so the profile does not depend on that choice—the unrestricted version is Table 15. The no-op arm—the receiver’s own states written back into the receiver—is reported wherever it was run and must equal the receiver’s baseline; it does, at every layer.

	layer	accuracy	CI ₉₅	fraction of the document’s own effect	
	0	0.875	[0.713, 1.000]	+1.00	(no-op arm 0.000)
	4	0.938	[0.819, 1.000]	+1.07	
	8	1.000	[1.000, 1.000]	+1.14	
	12	0.875	[0.713, 1.000]	+1.00	(no-op arm 0.000)
	13	0.625	[0.388, 0.862]	+0.71	(no-op arm 0.000)
	14	0.688	[0.460, 0.915]	+0.79	(no-op arm 0.000)
	15	0.125	[0.000, 0.287]	+0.14	(no-op arm 0.000)
	16	0.125	[0.000, 0.287]	+0.14	(no-op arm 0.000)
	18	0.062	[0.000, 0.181]	+0.07	(no-op arm 0.000)
	20	0.250	[0.038, 0.462]	+0.29	
	24	0.000	[0.000, 0.000]	+0.00	
	30	0.000	[0.000, 0.000]	+0.00	(no-op arm 0.000)
	every layer	0.875	[0.713, 1.000]	+1.00	
	<i>the filler document, unpatched</i>	0.000		0.00	
	<i>the gold document in context</i>	0.875		1.00	

C.15 WHAT THE CARRYING VECTOR IS MADE OF, IN FULL

The vector at the document’s span reproduces the document’s whole effect, so what it is composed of is answerable rather than speculative. Four decompositions, all at layer 8. **Unless marked “after the fix”, the numbers in this subsection come from the ten-calculator development runs made before the instrument fix**, read against the receiver’s 0.175 and the document’s 0.950; where a rerun exists it is given and takes precedence.

Table 9: Geometry of the content component over the document’s own span, by layer, Qwen3-8B, $n = 40$, all against the same fixed control document. Unlike the same quantity at the final prompt position, this vector is the size of the state it displaces. Two skills’ vectors are substantially aligned and most of each one’s energy lies along the all-skill mean—which, injected alone, recovers 0.11 here and 0.06 with the document after the question (both pre-fix) (§C.15). Participation ratios are centred, so they count the directions the span’s positions *differ* in. Cosines between content vectors depend on which control was subtracted: against each skill’s own paired control the cross-skill cosine reads 0.10 instead of 0.54, which is a fact about the control and not about the skills.

layer	$\ d\ $	$\cos(d_i, \bar{d})$	energy along $\bar{d}$	$\ d\ /\ h\ $	PR(d)	PR(h_{skill})	$\cos(d_i, d_k)$, other skill
0	10	0.720	0.54	0.91	81.8	50.5	0.483
4	28	0.738	0.56	0.94	77.1	45.7	0.511
8	53	0.755	0.58	1.04	94.8	63.8	0.539
12	68	0.766	0.60	1.09	64.2	46.3	0.560
16	106	0.759	0.59	1.19	17.1	13.9	0.556
20	153	0.755	0.58	1.15	17.5	15.2	0.547
24	266	0.750	0.57	1.06	17.2	14.8	0.535
28	455	0.756	0.58	1.08	14.4	13.1	0.547
32	751	0.753	0.58	0.98	17.5	13.8	0.541

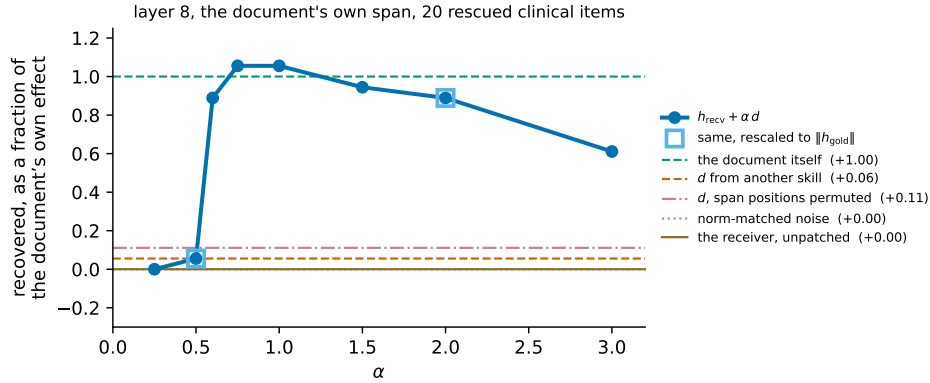


Figure 3: Dose response of the content injection over the document’s own span, layer 8, with its controls. Open squares are the same dose rescaled per position to the with-document norm; they coincide with the plain arms. Before the instrument fix; the post-fix dose sweep and norm comparison are in Appendix C.15. The horizontal lines are the three controls and the two baselines.

Not a common shift. Averaging d over *every* skill in the set and writing that average gives $\rho = 0.11$, against 1.06 for the document’s own. What makes this sharp rather than merely negative is how much of the vector that average is: measured against the same fixed control, two skills’ content vectors have cosine 0.48–0.56 at every depth, the cosine to the all-skill mean is 0.72–0.76, and 54–60% of each d ’s energy lies along that mean (Table 9). *More than half of the content vector points in a direction that carries nothing.* This is the span-level counterpart of g ’s flat 0.062 at the final position in §3: presence is not what acts, in either place.

Not a magnitude. The informative test is the dose curve against its norm-matched twin. At $\alpha = 2$ the patched state is $2h_{\text{skill}} - h_{\text{recv}}$, which is $1.76\times$ as long as the with-document state at layer 8; rescaling it per position back to $\|h_{\text{skill}}\|$ is a 43% change and costs nothing— $\rho = 0.89$ either way. At $\alpha = 0.5$ the state is $0.88\times$ and rescaling is likewise free (0.06 both).

We also ran the reverse, rescaling the with-document states down to the *receiver*’s norm, and it changes nothing ($\rho = 1.06$)—but that arm is uninformative and we report it only to say so: the two norms differ by 3.5% at layer 8, so it is barely a manipulation. The $\alpha = 2$ comparison is the one that carries the claim. The dose curve and its norm-matched twin coincide at every α we ran (0.5: 0.350 vs 0.350; 2: 0.875 vs 0.875; 3: 0.675 vs 0.650). The document’s effect is carried by the

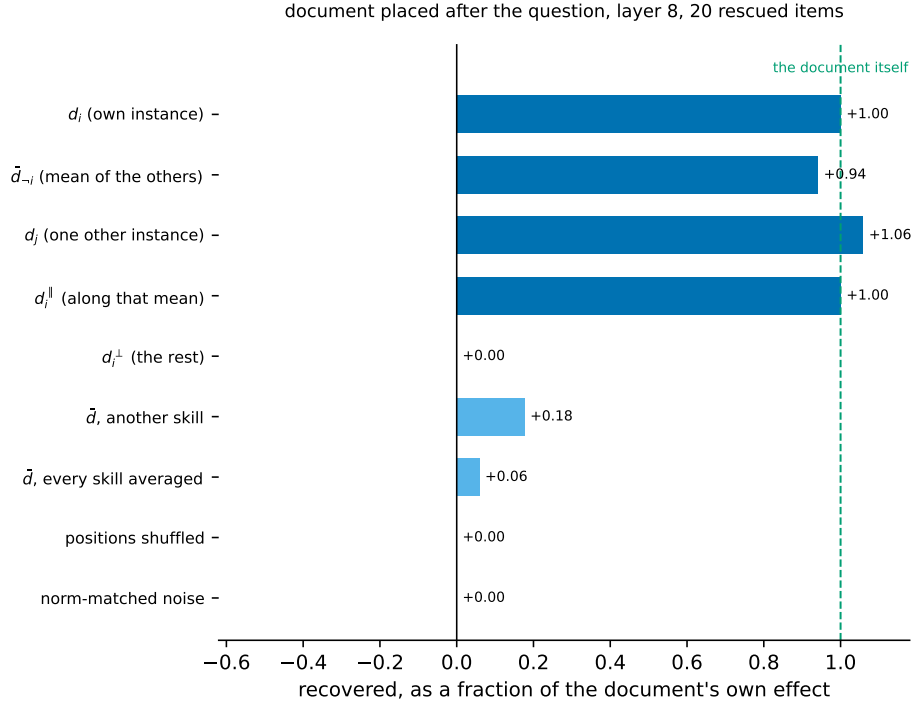


Figure 4: What the content vector contains once it is able to contain the task. Document placed after the question, layer 8, Qwen3-8B, the 40 rescued items, plotted as the fraction ρ of Eq. 2, before the instrument fix (post-fix numbers in Appendix C.15). Exact accuracies and intervals are in Table 18, Appendix D.

direction its span states point in and not by how far along it they go. Norm-matched Gaussian noise, correspondingly, gives 0.100, below the receiver. The norm arms were not rerun after the fix. (The arm listed next to the transplant in Table 1, 0.73, is not a norm manipulation: it is the transfer test’s paired control, the document’s own d written over only the prefix another skill’s document shares.)

Graded in α , and the transition is a step. After the fix, on the ten development calculators ($n = 16$), $\alpha = 0.4, 0.5, 0.55, 0.6, 0.7, 1, 2$ give $\rho = 0.06, 0.00, 0.06, 0.62, 0.94, 1.00, 0.81$: at the receiver up to 0.55 and at the document by 0.7. Before the fix the same sweep read 0.17, 0.03, 0.17, 0.88, 0.97, 1.00 ($n = 30$ under a looser restriction), and an earlier receiver gave $-0.14, +1.03, +1.03, +0.86$ at $\alpha = 0.5, 0.75, 1, 2$ (Figure 3, pre-fix): half the content is no better than none of it, while three quarters is already most of the effect. Past $\alpha = 1$ the recovery decays slowly rather than breaking, and against the other receiver, where the fine grid was run, it decays the same way with the norm held fixed ($\alpha = 2$: 0.875 plain against 0.875 rescaled; $\alpha = 3$: 0.675 against 0.650), so what overshooting costs is not the size of the state.

Localised, and in the section one would guess. Writing d over only part of the span localises the effect within the document without changing a single prompt token—the length, the positions and the rotary phases are all untouched, and only which states are overwritten changes. By halves (pre-fix), the first gives $\rho = 1.06$ and the second 0.28, and the first half agrees with the whole span item by item on 18 of 20. By quarters:

quarter of the span	n	1st	2nd	3rd	4th
forty calculators, after the fix	60	0.12	0.56	0.12	0.12
ten calculators, after the fix	16	0.12	0.81	0.06	0.06
ten calculators, before the fix	20	0.28	0.83	0.11	0.44
controls on the same forty (Table 1)	60	half dose	0.06, permuted	0.08, noise	0.10

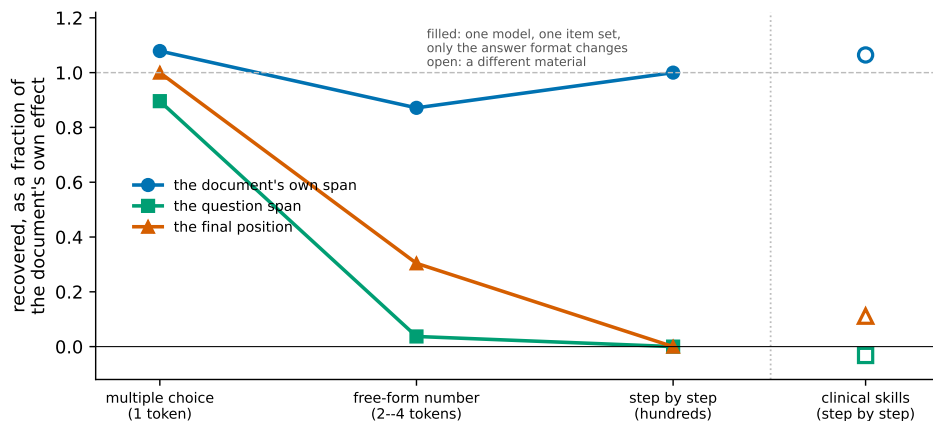


Figure 5: What each transplant channel recovers, as a fraction of what the document itself is worth in that setting, against how many tokens the answer occupies. Filled markers joined by lines are one model, one item set and one document with only the answer format changed; the clinical results are open and deliberately not joined, because that material changes the skill, the task and the prompt length at once. Plotting fractions rather than raw accuracies is necessary here: the four settings have donors at 0.550, 0.467, 1.000 and 0.950, and comparing raw numbers across them is how an earlier draft came to report that a *longer* answer made the document-span channel stronger. The document’s own span is flat at 1.00 across every answer format. The other two channels collapse as the answer lengthens, and the clinical points land where the step-by-step column says they should.

These documents make that interpretable, because all ten have the same section order: title, *Required Inputs*, *Computation* (sometimes preceded by a *Unit Conversion*), *Calculation Tool*, *Example*. Measured in the units the intervention actually uses—token positions within the span, not characters—the computational section’s *heading* falls at 0.18–0.42 of the span, *Calculation Tool* at 0.47–0.70, and *Example* at 0.61–0.79. So the second quarter is the *body* of the computational section, the formula itself, for every one of the ten; the first quarter is the title, the input list and at most the opening of that section; the third is a Python function signature and the start of a worked example; the fourth is the rest of the example.

Read that way the procedural core carries most of the effect, and every other quarter alone is at the control floor. Two things changed with the fix and the larger set. The worked example, which read 0.44 before the fix, reads 0.06 and 0.12 after it: *we withdraw the earlier statement that it carries about half as much as the formula*. And the formula alone, 0.81 on ten calculators, is 0.56 on forty against the whole document’s 0.87, so on the larger set the other three quarters matter in combination although none matters alone. The boundaries are approximate and vary by calculator, so we report the quarters and the section map rather than claiming an exact alignment.

Permuting the span’s own positions, which preserves the multiset of vectors and destroys only the correspondence between them and the tokens, gives 0.08 on forty calculators after the fix (0.21 before it): nearly all of the effect is in that correspondence.

Task information, when the vector is allowed to have any. Under the prompt format used everywhere else in this paper the question does not arise: the document precedes the question, so its states are a function of the document alone and we verified that they agree across a skill’s instances (bit for bit at equal prompt length, cosine ≥ 0.997 otherwise). To ask what happens when the vector *can* carry the task, we move the document behind the question, keeping the answer-format instruction last, and pad each instance’s question with filler tokens so that the document begins at the same absolute index for every instance of a skill—without which a donor from another instance would be written at the wrong rotary phase, the error described in §4. The format costs almost nothing behaviourally (the document is worth $0.025 \rightarrow 0.925$ instead of $0.025 \rightarrow 0.950$) and it raises the receiver’s own baseline to 0.350.

In that format the vector *can* carry the task, and at layer 8 it does not use it (read, like everything else here, on the calculators a wrong document never rescues). After the fix ($n = 28$, seven calculators) the instance’s own vector recovers $\rho = 1.00$, the mean of the skill’s other instances’ vectors 1.00, a single other instance’s vector 0.96, and the orthogonal remainder—everything specific to this patient note—0.04. Before the fix (Figure 4) the same arms read 1.00, 1.00, 1.06 and 0.00. At layer 16 the picture changes: the instance’s own vector still recovers 0.85, but the mean of the other instances’ vectors 0.63 and a single other instance’s 0.48, while the remainder alone stays at 0.04. By layer 16 the doc-last span holds something of the question that other instances’ vectors lack and the remainder cannot supply on its own.

The geometry agrees and says why, before any injection: the cosine between two instances’ d in this format is 0.998 at layer 0, 0.984 at layer 8 and 0.914 by layer 32, with the fraction of energy along the instance-mean falling from 0.997 to 0.880. At layer 8 the vector is 98% shared, so there is almost no instance-specific part to use; whether the remaining 12% at layer 32 ever matters is a question we can pose but have not answered.

Everything else replicates across the two prompt orders (pre-fix). Another skill’s vector recovers 0.18, the mean over every skill 0.06, the same vectors with the span’s positions permuted 0.00 and norm-matched noise 0.00; rescaling to the receiver’s own norm costs nothing (0.99); and the first half of the span carries 1.05 against the second half’s 0.43, the same concentration in the formula that the document-first format shows.

Where the document sits changes how deep the handover reaches. Placing the document after the question moves the cliff. We first reported this as a confirmed rank prediction, on the grounds that the doc-last span’s participation ratio past layer 16 read 28–33 against the doc-first format’s 14–17; §5.1 shows that both of those numbers are massive activations rather than representations, and once they are excluded the two formats have the same, flat dimensionality. The behavioural difference is real and the rank explanation for it is withdrawn.

The difference is large. On the same items, before the fix, as a fraction of each format’s own range:

layer	0	4	8	12	16
document first	1.00	1.07	1.14	1.00	0.14
document last	0.99	1.11	1.00	0.93	0.80

Both formats carry the document’s whole effect through layer 12. At layer 16 the document-first format carries a seventh of it and the document-last format four fifths; the latter’s own cliff comes later, between layers 16 and 20 (0.82 to 0.35). Under causal masking a document placed first cannot see the question and a document placed last can, so the doc-last span’s states are already question-conditioned when they are captured—which is the obvious candidate explanation, and is not one we have tested.

After the fix the layer-16 contrast holds: 0.85 with the document last ($n = 28$) against 0.19 with it first on forty calculators. Before the fix, at layers 0, 12, 16, 20 in the document-last format the mean of the skill’s other instances tracked the instance’s own (1.06/1.00, 0.88/0.94, 0.82/0.82, 0.29/0.35) with the remainder at the receiver; the rerun at layer 16 (0.63 against 0.85, above) does not reproduce that agreement, so we no longer claim that the instance-specific part never matters at depth—only that it does nothing alone.

What this rules out is the reading in which the handover depth is a fixed property of the network: it moves under a change to the prompt that leaves the model, the document and the items alone.

C.16 RELATED WORK, AT LENGTH

Skill benchmarks measure the effect, not the mechanism. A 2026 literature evaluates skill documents behaviourally: across domains (Li et al., 2026a), on executable software-engineering tasks (Han et al., 2026), and under different delivery protocols (Li et al., 2026b); a parallel line studies which skill to retrieve. Closest in motivation is distilling a skill into weights (Zhang & Qi, 2026), which asks what a skill *could* become rather than what it already is at inference time. None of this

work looks inside the forward pass, and the effect sizes it reports are exactly what our §7 shows can be format-dependent.

Task and function vectors, and the two limits on what one carries. That a context can be compressed into a single activation-space direction is established for in-context demonstrations (Hendel et al., 2023; Todd et al., 2024) and, in weight space, for fine-tuning (Ilharco et al., 2023). Methodologically we are in this line: our d is extracted and re-injected the same way. Two limits on that move are now known, and they are not the same limit.

Dong et al. (2026) prove the first. In a linear transformer, injecting one task vector is functionally a rank-one map, so a task whose input–output relation is high-rank—a bijection, for instance—cannot be encoded by a single vector; they confirm it on Llama-7B/13B and repair it by injecting several vectors at several demonstration positions. That result *explains one of ours*: on the synthetic tier, $\bar{d}$ —one vector shared by the whole item set, which is what a reusable skill vector would be—recovers 0.062, and the mapping it would have to encode is a 39-row lookup table. We do not claim that null as new; we attribute it.

The limit this paper is about is different, and the two can be separated empirically. On the same synthetic tier, the *item’s own* d_i recovers 0.812 through the same single position despite that same high-rank mapping, because the answer is one token; and on clinical skills the same injection recovers nothing because the answer is hundreds of tokens. Rank constrains which mappings one vector can express; length constrains how much of a decision any small position group can carry, whatever the mapping. Dura et al. (2026) state the length limit as a motivation for patching CoT sequentially rather than at a static position; §3.1 turns it into a controlled measurement, and §5.2 measures the rank of the *representation* rather than of the mapping.

We also differ in position. Function and task vectors are read from, and written to, one position, and are validated on tasks whose answer is a single token. Mehrafarin et al. (2026) show how far that can be pushed and, read against our result, show what the position holds: patching a single late position with a state taken from the model’s *own completed* chain of thought recovers the correct GSM8K answer at least 99% of the time, while our donor—taken from the prompt, before any reasoning—recovers nothing. Same position, same intervention; what differs is whether the answer already exists in the donor. Da Silva et al. (2025) report a reliability failure of the same channel from another direction: across 36 models, function vectors with default parameters recover 5-shot performance in 20% of model–task pairs.

When one vector is enough, and what that tells us. The strongest recent positive result for single-vector compression is Ardoin et al. (2026), who learn a weighted sum of an instruction prompt’s activations at a middle layer, inject the resulting *single* vector at one placeholder position in an early layer, and lose under two points of accuracy. We read it as convergent rather than contradictory, for two reasons, and both are load-bearing here. First, every task on which it succeeds has a short answer: capitals, ISO codes, continents, years, word translation, antonyms, currency codes—all single-word exact match—and ARC-Easy, which is a single letter. That is exactly the half of the answer-length axis where our own single-position patch also works (0.812 on the synthetic tier, §3.1); the half where it fails is the one their evaluation does not enter. Second, their optimal configuration extracts at a middle layer and injects at an *early* one—their stated reason is that an early injection leaves more layers to integrate it—which is independently the direction of our depth result (§5.1). Taken with Dong et al. (2026) and Wiegrefe et al. (2025), the three give one sentence: a single vector is a *selector* among behaviours the model already has—their appendix encodes nine tasks in one vector, about three bits—and not a carrier of content that needs a hundred directions to rebuild.

Three different objects are called “rank” here. Because this paper adds a fourth, we separate them explicitly. Dong et al. (2026) bound the rank of the *task map*: which input–output relations one injected vector can express. Sarkar & Deka (2026) measure the rank of the *Jacobian*, $\text{eff-rank } \mathbb{E}_x[J_L^T J_L]$ —how many directions of a layer causally reach the output—and report it to be layer-intrinsic and, notably, unchanged between Gemma-2-2B and Gemma-2-9B. Fernando & Guitchounts (2026) measure the rank of the *composed operator* across depth and find it collapses by about $65\times$ in Llama-3.1-8B. We measure the rank of a *representation*: the matrix $d \in \mathbb{R}^{m \times d}$ that one document contributes across its own m span positions. The distinction matters for what our re-

sult means. An earlier draft combined a layer-intrinsic causal capacity (Sarkar & Deka, 2026) with the participation ratio of d falling from 64 to 17 to argue that the handover closes on the *message* side; that fall is massive activations (§5.1), and with them excluded the ratio is flat, so we withdraw the argument and do not know on which side the window closes. We are also, among these four, the only one to truncate the injected object itself and read out behaviour; the other three use a theorem, a first-order attribution proxy, and a spectral surgery on the weights.

A caution applies to all of the descriptive ones, ours included. Sun et al. (2024) document activations three orders of magnitude above the median, carried by a fixed pair of hidden dimensions at a small set of token positions, appearing at a single layer (Shi et al., 2026) and acting as bias terms. A spectral statistic computed on a residual stream past that layer is a statistic about those activations. Our own centred participation ratio reads 1.0 over four hundred positions at layer 16 of Qwen3-8B and 66 once four positions are dropped (§5.1); an uncentred entropy rank, which is what Fernando & Guitchounts (2026) report collapsing by $65\times$, is if anything more exposed, since the massive component is not removed by centring there either. We do not claim their result is an artefact—their object is the composed operator, not a representation—but the check is cheap and we think it should be standard.

Compressing a document into parameters or into few tokens. A parallel line replaces the document with weights rather than activations—hypernetworks that emit LoRA parameters from a document (Charakorn et al., 2026), and context distillation into a knowledge module in both output and hidden space (Caccia et al., 2025). These are orthogonal to us in method (weights, and training, versus activations and none) but share the premise that a document’s effect is compressible. Wang et al. (2026) report that re-supplying the context after such internalisation *degrades* accuracy, which is a phenomenon our $t = g + d$ split gives a vocabulary for: what is internalised is not the same object as what the document contributes while present.

Activation patching. We follow the methodology literature on what patching licenses (Zhang & Nanda, 2024; Heimersheim & Nanda, 2024): transplants establish sufficiency, knockouts necessity, and neither alone establishes a route. Two recent results are close to ours and we read them as convergent evidence rather than as competition. Pola & Balasubramanian (2025) patch between minimal-contrast prompts differing only in an instruction span and find an “onset” layer past which interventions stop mattering; our layer-21 result—beyond which masking the document is free—is the same shape of finding from the opposite direction (knockout rather than patch) with a document rather than an instruction. Li et al. (2025) argue that transferable task representations are temporally local and assembled just in time; our knockouts say the same thing about reading a document, in that no single position group is necessary. Abdelsalam et al. (2026) likewise localise recoverable causal signal to a changed cue region plus a few intermediate tokens before aggregation near the output. Mechanistic work on retrieval-augmented generation (van Dort & Heuss, 2026) patches the residual stream to localise attribution but does not separate presence from content.

C.17 MULTIPLE CHOICE MANUFACTURES A FLOOR AND MERGES RETRIEVAL WITH EXECUTION

Our synthetic items exist in two forms with identical content: four-option multiple choice, and free-form (“give the number only”). The forms do not measure the same thing, and there is a mechanism in the literature for why. Wiegrefe et al. (2025) localise the choice of an answer *symbol* to a single middle layer and one to four attention heads per layer: in multiple choice the decision is assembled into one position and one token. That is exactly the regime in which a single-position patch transports the whole task, and it is why our final-position channel reads 1.000 there and 0.000 under a chain of thought on the same items.

Inspecting what the small model writes explains the top half. Without the document it echoes the quantity from the question (“2”); with the document it writes a *conversion factor from the table* (“21”, “180”, “60”). It retrieves and does not execute. Multiple choice scores retrieval as success whenever the retrieved number can be matched against an option, and the positional prior supplies a floor of 0.18–0.29 on top.

The bottom half is the sharper version, and it is a prediction we registered before running: if the multiple-choice form is solvable from a document’s structure rather than its numbers, a table with

every factor replaced must score near zero in free-form. It scores exactly zero. Scrambling the line order costs little in either format, which is what it should—our shuffle permutes lines and each row of the table is one line, so every factor is still there to retrieve, just not in a readable order.

Two consequences. A benchmark comparing skill effects across model scales in multiple-choice format is measuring a mixture whose composition changes with scale. And an ablation that perturbs a document and reports the accuracy drop is measuring, in multiple choice, mostly whether the document is still recognisably a document.

C.18 THE DECOMPOSITION ON THE SYNTHETIC TIER, IN FULL

The geometry at that position says the opposite of the behaviour. Before intervening it is worth seeing why the obvious measurement cannot settle this. The cosine between the displacement a skill causes at the last prompt position and the one an unrelated document of the same length causes is 0.93 at layer 0, 0.9475 at layer 4 and 0.91 at layer 10 on the synthetic control, falling to 0.35–0.41 only from layer 18—that is, only after the model has stopped reading. Read alone that says the injection is not content-specific; read against Table 7 it cannot be, because behaviour is all content. The decomposition resolves it: over the first half of the network $\|d\|$ is 0.38–0.45 of $\|t\|$, so most of what such a cosine measures is shared, and through layer 16 $\|g\|$ exceeds $\|t\|$ —the wrong document moves the state *further* than the right one, and the two components partly cancel. Appendix B gives the full picture. We use geometry only to choose what to intervene on, and then intervene.

We add each component back to the no-document state at the last prompt position and let the forward pass continue: $h' = h_{\text{no}} + \alpha v$ for $v \in \{d_i, g_i, t_i, \bar{d}, d_j\}$. The upper reference is replacing the state outright with h_{skill} ; the lower reference is no intervention.

Figure 6 is the causal counterpart of that geometry (numbers in Table 23, Appendix D). Four of its curves are worth reading separately.

The presence component repairs nothing—0.062, three items of 48, at every layer past 20, and 0.00–0.02 before. It is not inert noise: injected on its own it takes whole-set accuracy from the no-document baseline of 0.180 to 0.145, which is the accuracy of actually putting the wrong document in the prompt. On the rescued cell it gives 0.062, and the wrong document in context also gives 0.062. It faithfully reproduces the behaviour of having a document present, and that behaviour is worse than having none.

The content component repairs most of the cell—0.812 at layer 26 against an upper reference of 1.000, having been dismissed by every geometric measurement in §B as a small residual.

At one patched position it is under-scaled, not incomplete. The dose-response is monotone and steep— $\alpha = 0.5, 1, 2$ give 0.31, 0.81, 0.98—so the missing 0.19 at $\alpha = 1$ is a magnitude deficit along a correct direction, which is expected: $h_{\text{wrong}} + d_i$ is exactly h_{skill} , while adding d_i to h_{no} lands elsewhere. It does not generalise to more positions: at $k = 16$ in free-form the dose is flat between $\alpha = 1$ and 2 (0.232, 0.268) and never approaches the full replacement’s 0.554. The two components are additive by construction and not additive in effect. What survives every configuration is the asymmetry: g alone is inert at every k , layer and α .

The receiver does not matter, and a single shared vector is not enough. Injecting h_{skill} into the prompt that already holds the *wrong* document gives 0.979–1.000, the same as into a prompt with no document—whatever the surrounding 700 tokens contribute, the final position does not need it. But $\bar{d}$, one vector for the whole item set, which is what “a reusable skill vector” would mean, recovers 0.042–0.146 from layer 21 against 0.479–0.812 for the item’s own. Splitting d_i along $u = \bar{d}/\|\bar{d}\|$ and orthogonal to it and injecting each half separates them and reverses the answer at layer 21: past it the parallel half carries ≈ 0.9 of the norm and 0.05 of the effect while the orthogonal half carries 0.44 of the norm and 0.60 of the 0.81; before it the two swap, inside the band §7 independently identifies as one where the forward pass is disturbed.

The patch reproduces the document’s whole behavioural signature, not just its successes. Across all four cells, replacing the final position with h_{skill} gives 1.000/0.000/1.000/0.000 on rescued / persistent / kept / broken at layer 26 on the 1.7B, and 1.000/0.024/1.000/0.000 at layer 30 on the 8B—cell by cell, what the document itself does by the definition of the cells. It reproduces the skill’s failures and its damage as faithfully as its repairs. Adding d_i gives 0.812/0.112/1.000/0.125

and 0.864/0.024/1.000/0.077: the same signature, attenuated. A patch that only ever helped would be consistent with a generic “try harder” effect; one that also breaks exactly the items the document breaks is carrying the document’s decision. The presence component passes the matching test against *its* document: $h_{\text{no}} + g$ scores 0.062 on the rescued cell where the wrong document in context scores 0.062 (1.7B), and 0.250 where it scores 0.250 (8B), across four different control documents (Appendix A).

Which control you subtract chooses what you measure. d is defined against a control document, and the choice is a measurement decision rather than a detail. Against an *unrelated* document of the same length, d is topic, procedure and numbers together and repairs 0.812 of the rescued cell; against the *same* document with every number replaced it is the numbers alone and repairs 0.229–0.312 (layers 21–27); against the same document with its sentences shuffled, 0.188–0.292. The three differ by a factor of three to four, so a paper that reports “the content component” without naming its control has not specified the quantity. Appendix A gives the full comparison and the accompanying behavioural ablations.

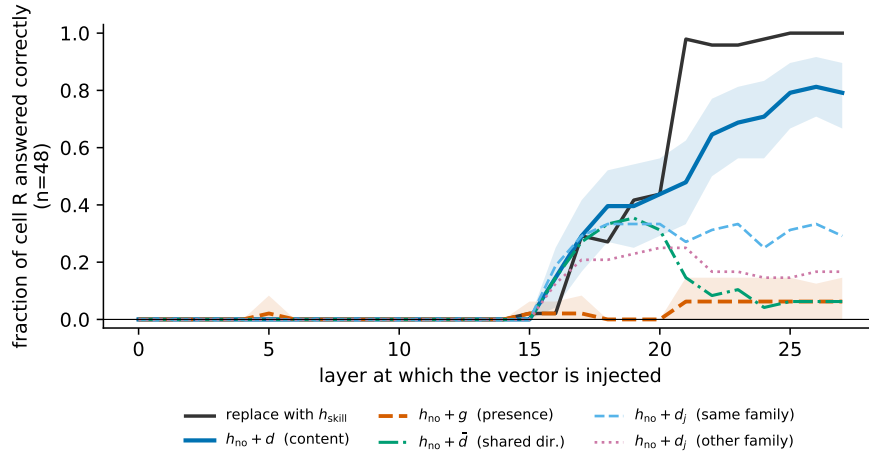


Figure 6: What each injected component repairs, on the cell the document rescues. Shaded bands are bootstrap 95% CIs for the two conditions the argument rests on. $h_{\text{no}} + t$ is omitted because it equals h_{skill} identically and its curve lies under the black one—we run it, and it agrees item by item. Synthetic control, multiple choice, Qwen3-1.7B, fp32, $n_R = 48$.

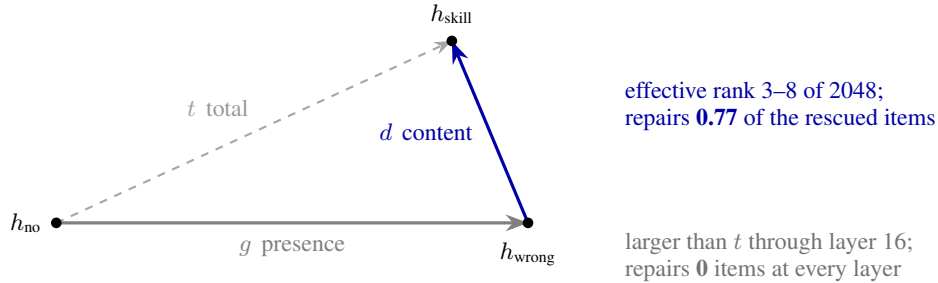


Figure 7: The decomposition, drawn to scale at layer 10 of the synthetic control ($\|g\| = 39.0$, $\|d\| = 16.3$, $\|t\| = 36.0$). A document of the right length and the wrong content already moves the final prompt position most of the way—in fact slightly further than the real skill does, which is why the content arrow points partly back. Everything this paper is about is the short blue arrow.

Table 10: The same items under two answer formats. Top: the effect itself. In multiple choice the small model shows a +12pp skill effect over an 0.18 floor; in free-form it shows nothing at all, over a floor of exactly zero, and the effect appears only at 8B. Bottom: the same four documents under both formats, where “share” is $(\text{acc} - \text{none})/(\text{gold} - \text{none})$. Destroying every number in the document keeps two thirds of the effect in multiple choice and none of it in free-form.

model	format	no document	wrong document	gold skill	effect
Qwen3-1.7B	multiple choice	0.180	0.145	0.300	+12.0pp
Qwen3-1.7B	free-form	0.000	—	0.000	0
Qwen3-8B	multiple choice	0.292	0.242	0.550	+25.8pp
Qwen3-8B	free-form	0.000	0.000	0.467	+46.7pp
document in context		multiple choice (1.7B)		free-form (8B)	
		acc	share	acc	share
none		0.180	—	0.000	—
same document, <i>factors</i> replaced		0.260	0.67	0.000	0.00
same document, <i>lines</i> scrambled		0.270	0.75	0.325	0.70
gold skill		0.300	1.00	0.467	1.00

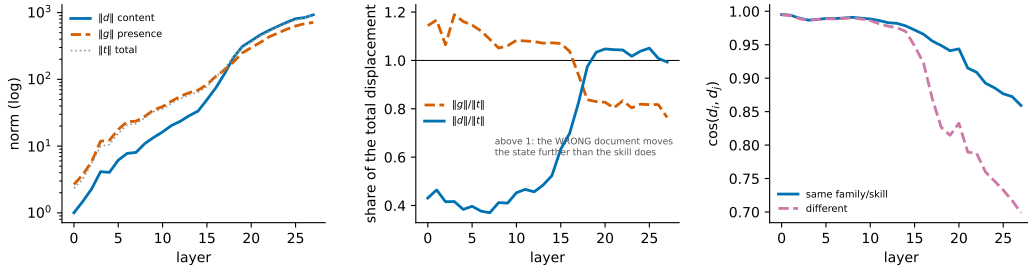


Figure 8: The decomposition across depth, synthetic control, Qwen3-1.7B, fp32, $n = 200$. Left: the three norms. Middle: each component as a share of the total displacement; above 1 the wrong document moves the state further than the skill does, which is the case until layer 17. Right: how aligned two items’ content vectors are, split by whether they use the same row of the conversion table.

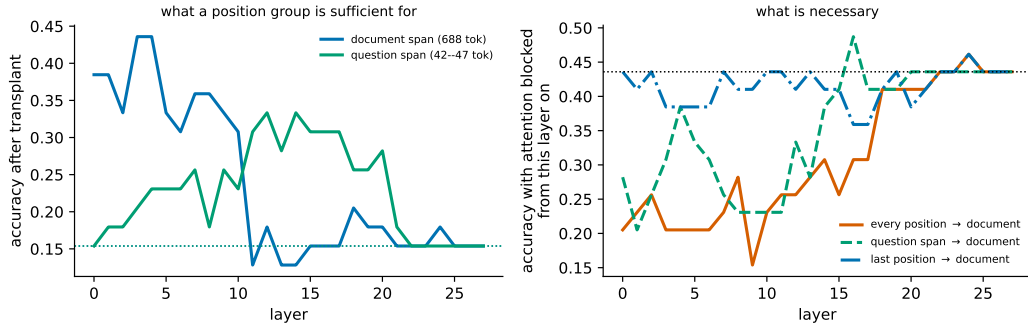


Figure 9: Left: what each position group is sufficient for, as a function of the layer at which its states are transplanted; dotted lines are each channel’s own receiver baseline. Right: what is necessary—accuracy when attention to the document is masked from the given layer to the end, cumulatively; the dotted line is the unmasked baseline. Synthetic control, multiple choice, Qwen3-1.7B, $n = 39$.

Table 11: Peak accuracy of each transplant channel, by model and answer format. “Donor” is the with-skill forward and “receiver” the prompt being patched into; the two span channels patch into a length-matched filler-substituted prompt, the final-position channel into the no-document prompt, which is why their baselines differ. Peaks in **bold** where the channel recovers most of the donor.

channel	configuration	recv.	peak (layers)	donor
document span (688 tok)	1.7B, multiple choice	0.154	0.436 (3–4)	0.436
	8B, multiple choice	0.233	0.575 (0–6)	0.550
	8B, free-form	0.008	0.408 (0 only)	0.467
question span (26–50 tok)	1.7B, multiple choice	0.154	0.333 (12–14)	0.436
	8B, multiple choice	0.233	0.517 (21)	0.550
	8B, free-form	0.008	0.025 (any, i.e. none)	0.467
final position(s)	1.7B, MC, rescued cell	0.000	1.000 (24–27)	1.000
	8B, MC, rescued cell	0.000	1.000 (30–35)	1.000
	8B, free-form, rescued cell	0.000	0.304 ($k=1$), 0.554 ($k=16$)	1.000

Table 12: Question-span transplants on the same material and construction. *Which* positions are transplanted is the whole difference between this table and Table 8.

transplanted positions	donor	L0–12	L16	L22	L30
question span (50–1697 pos., $n = 40$)	the gold skill	0.10–0.23	0.150	0.200	0.154
	the receiver itself (no-op)	0.13–0.20	0.175	0.125	0.154
no document / the gold skill in context		0.025	/	0.950	

D ADDITIONAL TABLES

Table 13: Where the aggregate log-probability gap comes from, synthetic control, layer 21, $n = 39$. A cell’s contribution to the mean is its size share times its own gap, so the column adds up exactly. The mean vector beats the true state overall because F and B—the cells where the document does not work—contribute 133% of the gap; on R, where it does work, the true state wins by 2.5 nats and by 0.69 in accuracy.

cell	n	mean – real (nats)	contribution to +2.747	acc. real	acc. mean
R rescued	13	−2.506	−0.835 (−30%)	0.923	0.231
F persistent	17	+5.907	+2.575 (+94%)	0.059	0.294
K kept	4	−0.740	−0.076 (−3%)	1.000	0.500
B broken	5	+8.455	+1.084 (+40%)	0.000	0.200

Table 14: The numbers behind Figure 1, restricted as the figure is to the calculators whose receiver never solves an item.

	accuracy	CI ₉₅
<i>the gold document in the prompt, and no document at all</i>		
gold document in context	0.950	[0.854, 1.000]
no document	0.050	[0.000, 0.146]
<i>written into the last prompt position (best layer of 10, 14, 20, 30 for each arm)</i>		
replace the state with h_{skill}	0.150	[0.000, 0.306]
$h + d$	0.150	[0.000, 0.306]
$h + g$	0.050	[0.000, 0.146]
$h + d_j$, another instance of this skill	0.150	[0.000, 0.306]
$h + d_j$, another skill	0.100	[0.000, 0.231]
<i>written over the document’s own span, layer 8</i>		
the filler document, unpatched (the floor here)	0.000	[0.000, 0.000]
$h_{\text{recv}} + d$ ($= h_{\text{skill}}$, $\alpha = 1$)	1.000	[1.000, 1.000]
$h_{\text{recv}} + 2d$	0.812	[0.621, 1.000]
$h_{\text{recv}} + \frac{1}{2}d$	0.000	[0.000, 0.000]
$h_{\text{recv}} + \bar{d}$, shared prefix only	0.938	[0.819, 1.000]
$h_{\text{recv}} + \bar{d}_j$, another skill, same prefix	0.188	[0.000, 0.379]
$h_{\text{recv}} + d$, span positions permuted	0.000	[0.000, 0.000]
$h_{\text{recv}} + \text{noise}$, $\ \cdot\ $ matched per position	0.000	[0.000, 0.000]
h_{recv} itself (must be a no-op)	0.000	[0.000, 0.000]

E EXPERIMENTAL DETAIL

Models and numerics. Qwen3-1.7B (28 layers, $d = 2048$) and Qwen3-8B (36 layers, $d = 4096$); §5.3 adds Qwen3-0.6B (28 layers, $d = 1024$) and Mistral-7B-Instruct-v0.3 (32 layers, $d = 4096$), whose chat template has no system role, so the system text is merged into the user turn. Greedy decoding throughout, `enable_thinking` disabled and verified by a self-test on the rendered template. Eager attention wherever an attention mask is manipulated, because sdpa and flash silently ignore a 4D mask; layer-output hooks are implementation-independent, so the decode-heavy runs use sdpa, which is about $2.5\times$ faster on a 2k-token prompt. The 1.7B synthetic runs are fp32: bf16 costs 0.5–2.5 nats on the log-probability of a token deep in the tail, and every log-probability quantity here is a difference of such numbers. The 8B runs are bf16 and the gaps we report from them are 5–8 nats, well outside that error; the accuracy channel is immune to it either way.

Self-tests, and one that failed. Three checks run with the experiments. (i) *Self-transplant*: donor and receiver are the same forward, so the patch must be an exact no-op. On the fp32 synthetic runs the deviation is 0.0 nats at every layer. On the bf16 8B runs it is up to 0.38 nats—bf16 arithmetic over two sequence lengths—while the accuracy channel remains an exact no-op at every layer, which is how we tell the two apart. (ii) *Last-layer patch*: replacing the final layer’s output at the patched

Table 15: The document-span depth sweep over all forty items, unrestricted. Compare Table 8: the profile is the same shape with a higher floor, because the calculators this adds are ones a wrong document already rescues.

layer	accuracy	CI ₉₅	fraction of the document’s own effect	
0	0.950	[0.882, 1.000]	+1.00	(no-op arm 0.150)
4	0.950	[0.882, 1.000]	+1.00	
8	1.000	[1.000, 1.000]	+1.06	
12	0.875	[0.773, 0.977]	+0.90	(no-op arm 0.150)
13	0.675	[0.530, 0.820]	+0.65	(no-op arm 0.200)
14	0.700	[0.558, 0.842]	+0.68	(no-op arm 0.175)
15	0.400	[0.248, 0.552]	+0.29	(no-op arm 0.150)
16	0.400	[0.248, 0.552]	+0.29	(no-op arm 0.175)
18	0.250	[0.116, 0.384]	+0.10	(no-op arm 0.175)
20	0.475	[0.320, 0.630]	+0.39	
24	0.175	[0.057, 0.293]	+0.00	
30	0.150	[0.039, 0.261]	−0.03	(no-op arm 0.150)
every layer	0.950	[0.882, 1.000]	+1.00	
the filler document, unpatched		0.175	0.00	
the gold document in context		0.950	1.00	

Table 16: The same depth sweep run a second time against the other receiver—a single fixed wrong skill rather than each skill’s own paired control—on the same items. The two agree wherever the first is well behaved; the one bump in the first run does not replicate.

layer	accuracy	CI ₉₅	fraction of the document’s own effect	
0	0.950	[0.882, 1.000]	+1.00	(no-op arm 0.125)
4	1.000	[1.000, 1.000]	+1.06	
8	0.975	[0.927, 1.000]	+1.03	(no-op arm 0.125)
12	0.800	[0.676, 0.924]	+0.81	
16	0.375	[0.225, 0.525]	+0.28	
20	0.275	[0.137, 0.413]	+0.16	(no-op arm 0.150)
the filler document, unpatched		0.150	0.00	
the gold document in context		0.950	1.00	

position makes that position’s logits the donor’s by construction, so the next token must match exactly. (iii) *Query-side control* for the knockouts.

An error worth recording. The first version of the real-skill scoring code patched at index −1 of a forward run over prompt *and* answer, which is the last token of the answer—after every position whose logits are scored. Every arm returned the untouched baseline to six decimals, which is what made it visible. Relative indices are the wrong default whenever the forward length varies between the capture and the patch.

Control documents. The synthetic controls are generated from the skill document by `make_controls.py`, which checks two things: the corrupted document’s numbers must not coincide with the task set’s gold answers (7 of 249 do, 2.8%), and the shuffle must not be the identity. All three controls are stored under the same filename in separate directories, because the injector derives the “# Skill: *name*” header from the file stem—a file called `SKILL.zorb-units.corrupted.md` would announce the manipulation in the prompt, and the control would then differ from the treatment in two ways.

Item set. The synthetic set is generated from the conversion table with the value range 2–40, dropping any item whose answer also appears in a worked example of the document ($n = 358$; the sweeps use the first 120 or 200). Distractors are values reachable by misreading the table—the neighbouring row, the wrong family, the inverted direction—never random numbers, and the key is rotated across A/B/C/D so a position-biased model still scores at chance.

Table 17: The same arms over all forty items, unrestricted, after the instrument fix. The six calculators this adds are ones the model can partly do without the gold document: their receiver scores 0.375, another skill’s vector 0.667 and norm-matched noise 0.375, against 0.000, 0.188 and 0.000 on the four that need the document. Pooling them raises the transfer arm from $\rho = 0.19$ to 0.32, which is why the main text reports the restricted set.

	accuracy	CI ₉₅
<i>the gold document in the prompt, and no document at all</i>		
gold document in context	0.950	[0.854, 1.000]
no document	0.050	[0.000, 0.146]
<i>written into the last prompt position (best layer of 10, 14, 20, 30 for each arm)</i>		
replace the state with h_{skill}	0.150	[0.000, 0.306]
$h + d$	0.150	[0.000, 0.306]
$h + g$	0.050	[0.000, 0.146]
$h + d_j$, another instance of this skill	0.150	[0.000, 0.306]
$h + d_j$, another skill	0.100	[0.000, 0.231]
<i>written over the document’s own span, layer 8</i>		
the filler document, unpatched (the floor here)	0.225	[0.096, 0.354]
$h_{\text{recv}} + d$ ($= h_{\text{skill}}, \alpha = 1$)	1.000	[1.000, 1.000]
$h_{\text{recv}} + 2d$	0.800	[0.676, 0.924]
$h_{\text{recv}} + \frac{1}{2}d$	0.000	[0.000, 0.000]
$h_{\text{recv}} + \bar{d}$, shared prefix only	0.975	[0.927, 1.000]
$h_{\text{recv}} + \bar{d}_j$, another skill, same prefix	0.475	[0.320, 0.630]
$h_{\text{recv}} + \bar{d}$, span positions permuted	0.200	[0.076, 0.324]
$h_{\text{recv}} + \text{noise}$, $\ \cdot\ $ matched per position	0.225	[0.096, 0.354]
h_{recv} itself (must be a no-op)	0.225	[0.096, 0.354]

Table 18: What the content vector contains once it is able to contain the task—the numbers behind Figure 4.

donor written over the document’s span	accuracy	CI ₉₅	fraction
<i>the filler document, unpatched</i>	0.000		0.00
<i>the gold document in context</i>	0.850		1.00
d_i , this instance’s own	0.850	[0.694, 1.000]	+1.00
$\bar{d}_{-i}$, the mean over the skill’s other instances	0.800	[0.625, 0.975]	+0.94
d_j , one other instance	0.900	[0.769, 1.000]	+1.06
$d_i^{\parallel}$, the half along that mean	0.850	[0.694, 1.000]	+1.00
$d_i^{\perp}$, the rest	0.000	[0.000, 0.000]	+0.00
$\bar{d}$, another skill	0.150	[0.000, 0.306]	+0.18
$\bar{d}$, averaged over every skill	0.050	[0.000, 0.146]	+0.06
the same d_i , span positions permuted	0.000	[0.000, 0.000]	+0.00
noise, norm matched per position	0.000	[0.000, 0.000]	+0.00

Table 19: Three transplant channels, synthetic control, Qwen3-1.7B, $n = 39$. Each row is a separate experiment; the receiver baseline differs because the document-span and question-span transplants run inside a prompt that still contains a filler document.

channel	positions transplanted	effective layers	peak acc.	receiver baseline
document span	the document’s own 688 tokens	0–10	0.436 (L3–4)	0.154
question span	the question’s 42–47 tokens	11–16	0.333 (L12/14)	0.154
final position	the last prompt token	21–27	0.436–0.487	0.231

Table 20: The same measurements at two model sizes. The multiple-choice presence row is the reason we take the free-form format as the primary evidence: on the 8B, chance on four options is 0.250, and that is exactly where g sits.

	Qwen3-1.7B	Qwen3-8B
multiple choice: none / wrong doc / skill	0.180 / 0.145 / 0.300	0.292 / 0.242 / 0.550
free-form: none / wrong doc / skill	0.000 / — / 0.000	0.000 / 0.000 / 0.467
MC, rescued cell: replace outright	1.000	1.000
MC, rescued cell: $+d$	0.812	0.864
MC, rescued cell: $+g$	0.062	0.250 (= chance)
free-form, rescued cell: replace outright	—	0.304
free-form, rescued cell: $+d$	—	0.268
free-form, rescued cell: $+g$	—	0.018
layers where $\ g\ > \ t\ $	0–16	0–12
PR(d) centred, deepest layer	5.6	18.1
$\cos(d_i, d_j)$ same / other family, deepest	0.859 / 0.699	0.584 / 0.530

Table 21: The content component split along its own shared direction and orthogonal to it, rescued cell, $n_R = 48$. The crossover at layer 21 is the same layer at which cumulative attention knockout on the document becomes free (§5) and at which another item’s content vector stops working (§6).

injected	L16	L18	L20	L21	L24	L26
$h_{\text{no}} + d_i$ (whole)	0.146	0.396	0.438	0.479	0.708	0.812
$h_{\text{no}} + (d_i \cdot u)u$ (shared dir.)	0.146	0.354	0.292	0.125	0.042	0.062
$h_{\text{no}} + d_i^\perp$ (item-specific)	0.000	0.083	0.146	0.521	0.583	0.604
$h_{\text{no}} + \bar{d}$ (shared dir., fixed norm)	0.146	0.333	0.312	0.146	0.042	0.062

Table 22: The decomposition of Eq. 1, synthetic control, Qwen3-1.7B, fp32, $n = 200$, at the last prompt position. PR is the participation ratio of $\{d_i\}$ after removing their shared mean direction: the effective number of dimensions the item-to-item variation occupies, out of 2048. The last column is the fraction of a single item’s d_i that lies along that shared direction.

layer	$\ d\ $	$\ g\ $	$\ d\ /\ t\ $	$\cos(d, g)$	$\cos(d_i, d_j)$		PR(d)	on mean
					same fam.	cross fam.		
0	1.00	2.67	0.43	−0.50	0.995	0.995	2.6	0.997
10	16.31	39.02	0.45	−0.39	0.988	0.986	4.0	0.994
14	33.49	68.67	0.52	−0.38	0.978	0.970	8.3	0.987
21	462.22	355.62	1.05	−0.44	0.915	0.789	5.6	0.916
27	926.57	712.91	1.00	−0.38	0.859	0.699	5.6	0.874

Table 23: Fraction of the rescued cell answered correctly under each injected component. Synthetic control, multiple choice, Qwen3-1.7B, fp32, $n = 200$, rescued cell $n_R = 48$. Bootstrap 95% CIs over items are ± 0.12 to ± 0.15 at these magnitudes and are given in the appendix. $\bar{d}$ is the mean content direction over items, restored to the typical per-item norm; d_j is another item’s content vector, drawn from the same or a different conversion family. The row that licenses the name “content component” is g .

injected at the last prompt position	L17–20	L21	L24	L26	L27
h_{skill} (replace outright; upper ref.)	0.27–0.44	0.979	0.979	1.000	1.000
h_{skill} into the wrong-document prompt	0.29–0.75	0.979	0.979	1.000	1.000
$h_{\text{no}} + 0.5 d_i$	0.21–0.31	0.250	0.292	0.312	0.292
$h_{\text{no}} + d_i$ (content)	0.29–0.44	0.479	0.708	0.812	0.792
$h_{\text{no}} + 2 d_i$	0.29–0.63	0.875	0.979	0.979	0.958
$h_{\text{no}} + g_i$ (presence)	0.00–0.02	0.062	0.062	0.062	0.062
$h_{\text{no}} + \bar{d}$ (one shared direction)	0.27–0.35	0.146	0.042	0.062	0.062
$h_{\text{no}} + d_j$ (same family)	0.29–0.33	0.271	0.250	0.333	0.292
$h_{\text{no}} + d_j$ (other family)	0.21–0.25	0.250	0.146	0.167	0.167